

KCSE AGRICULTURE MODEL TEST PAPERS

Kenya Certificate of Secondary Education (K.C.S.E.)

FORM 4 AGRICULTURE

MODEL TEST PAPER 1 AGRICULTURE PAPER 1

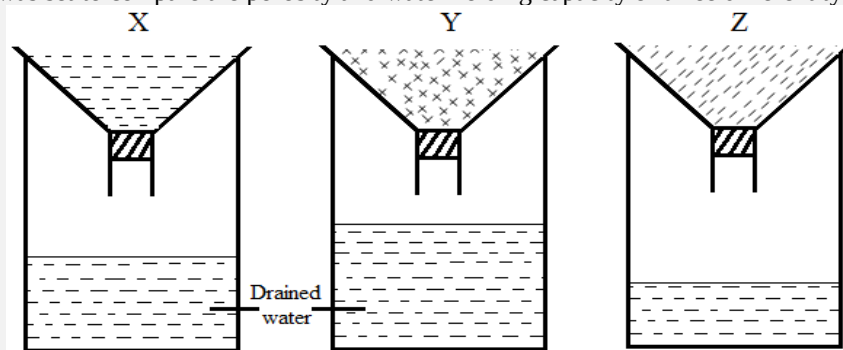
SECTION A (30 MARKS)

1. Differentiate between livestock farming and arable farming. (1mk)
2. State four characteristics of intensive farming. (2mks)
3. List two aspects of light that influence agriculture. (1mk)
4. State two biotic factors that are useful in agricultural production. (1mk)
5. Mention two conditions when opportunity cost is zero in a farm enterprise. (1mk)
6. Give four reasons why it is important to stake tomatoes. (2mks)
7. List four types of land reforms that have taken place in Kenya. (2mks)
8. Distinguish between seed dressing and seed inoculation. (1mk)
9. State two methods of utilizing sorghum as a forage crop. (1mk)
10. State what is meant by 'trap crop' in crop pest control. (1mk)
11. Give two benefits of border planting form of agroforestry to a farmer. (1mk)
12. Give four ways by which water pollution can be reduced or controlled. (2mks)
13. a) What is biological weed control? (1mk)
b) State two advantages of biological weed control. (1mk)
14. a) State two functions of sulphur in plants (1mk)
b) Give four signs that could enable a farmer to identify compost manure that is ready. (2mks)
15. List four farming practices that may help to achieve minimum tillage on a farm. (2mks)
16. Explain the following terms. (3mks)
a) Hardening-off
b) Seedling bed
c) Pricking out
17. Mention two sources of capital used by farmers. (1mk)
18. a) What is the meaning of the term 'utility' in farming? (1mk)
b) State four uses of gross margin analysis on a farm business. (2mks)

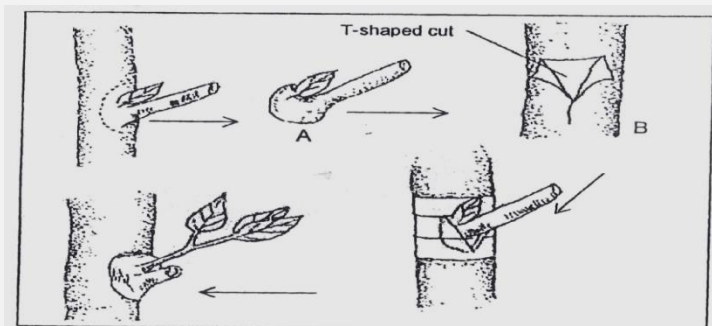
SECTION B (20 MARKS)

Answer all questions in this section

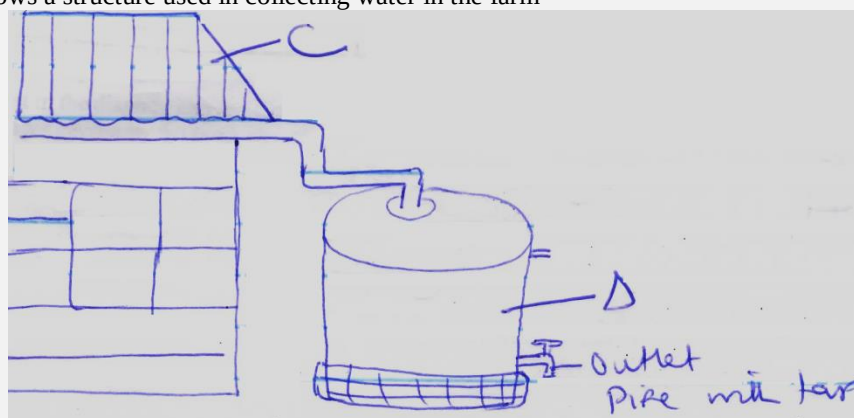
19. The experiment below was set to compare the porosity and water holding capacity of three different types of soils.



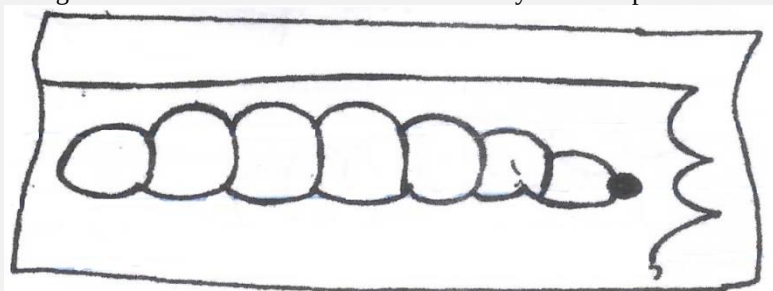
- i) Identify the soils in each of the following funnels labeled. X Y (1mk)
 - ii) Which of the type of soil can be said to have the highest porosity rate? (1mk)
 - iii) Give reasons for your answer. (1mk)
 - iv) Which type of soil would be suitable for planting paddy rice? (1mk)
 - v) Explain your answer in (iv) above. (1mk)
20. Use the diagram below to answer the questions that follow.



- Name the practice in the diagram above (1mk)
 - Identify parts A and B (2mks)
 - Give four advantage of the operation shown on the diagram above in crop production. (2mks)
21. The diagram below shows a structure used in collecting water in the farm



- Identify the method of collection above. (1mk)
 - Give three factors that will determine the amount of water collected in the structure labeled D. (1 ½ mks)
 - Outline two maintenance practices that one should carry out on part labeled C above. (1mk)
 - Name three other methods used in collecting water in the farm. (1 ½ mks)
22. The diagram below shows a maize stalk infected by a certain pest. Observe it and answer the questions that follow.



- Identify the pest in the diagram above. (1mk)
- Give two damages caused by the pests identified in (a) above. (2mks)
- Other than maize name other crops attacked by the pest (a) above. (1mk)
- Give two cultural measures for the pest (a) above. (1mk)

SECTION C

Answer two questions in this section.

23. a) A farmer has 30 hectares of arable land, 20 hectares of which is planted with maize and 10 hectares with grass hay. He wishes to replace 10 hectares of maize with irish potatoes next year. The fertilizer rate will have to be increased from 2 bags per hectare for maize to 4 bags per hectare of potatoes. As a result of the change, extra 40 men days of manual labour per hectare will be necessary at the rate of Kshs. 100 per man per day. The average yield per hectare of maize is 35 bags and for potatoes 120 bags. The price paid is Ksh. 1,000 per bag of maize and 600 per bag of potatoes. Maize seed cost Ksh. 750 per hectare and potatoes cost Kshs. 800 per hectare. Fertilizer costs are 650 per hectare.

i) Draw up a partial budget for the farmer.

(8mks)

ii) Advise the farmer accordingly.

(2mks)

b) Describe the importance of the following management practices in growing cabbages.

i) Top dressing

(2mks)

ii) Weed control

(2mks)

iii) Pest control

(2mks)

iv) Disease control

(2mks)

v) Harvesting

(2mks)

24. a) Explain ten effects of strong wind on crop production.

(10mks)

b) Describe ten cultural or biological methods of controlling soil erosion.

(10mks)

25. a) Explain five principles that should be considered when planning a crop rotation programme

(10mks)

b) i) Explain five factors that determines the time of harvesting.

(5mks)

ii) Explain five effects of liberalization of farming in Kenya.

(5mks)

AGRICULTURE PAPER 2

SECTION A (30 MARKS)

1. Give two reasons of flushing in sheep management

(1mk)

2. Give any two demerits of electricity as main source of power in the farm.

(1mk)

3. Which equipment is used to measure specific gravity of milk.

(1mk)

4. Name the breeding system by;

a) Mating a Friesian bull with Zebu cow.

(1mk)

b) Mating a Jersey cow from Kenya with a Jersey bull from Europe.

(1mk)

5. Name four breeds of dairy goats.

(2mks)

6. State two factors that could lead to failure to conceive in sows after service.

(1mk)

7. State four advantages of bucket feeding method in calves.

(2mks)

8. List four viral diseases of livestock.

(2mks)

9. Give two pairs of tools that must be used together in livestock production.

(2mks)

10. Outline four factors that determine the nutritional requirements of an animal.

(2mks)

11. Give the mineral deficiency of the following nutritional disorders in livestock nutrition.

(4mks)

i) Anaemia

ii) Osteomalacia

iii) Goitre

iv) Curled toe paralysis

12. Give four practices carried out on fish before preservation.

(2mks)

13. Name the parts of the digestive system of poultry that act as the stomach.

(1mk)

14. Differentiate between the following terms.

(4mks)

i) Digestion and digestibility

ii) Forage crops and fodder crops

15. Which hormone stimulates milk let down in cows?

(1mk)

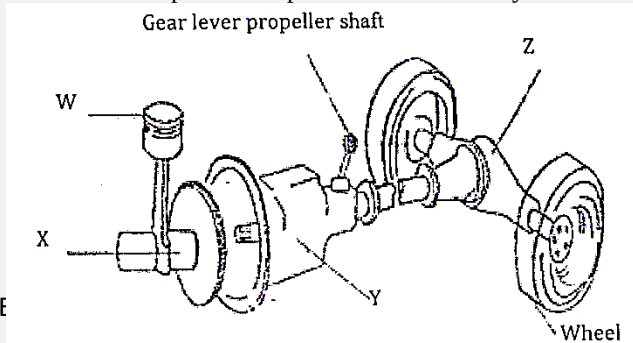
16. Distinguish between castration and caponisation

(2mks)

SECTION B

Answer all questions in this section.

17. The diagram below shows part of the power transmission system of a tractor. Study it carefully and answer the questions below.



GOLDEN E

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Name the parts labeled W and Y

(1mks)

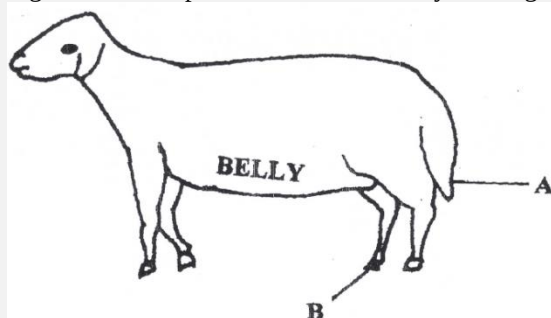
a) State the functions of the parts X and Z

(2mks)

b) Name two ways in which the power is transmitted from the engine.

(2mks)

18. A diagram of a sheep is shown below. Study the diagram and answer the questions that follow.



i) What operation is usually carried out on the part labeled A?

(1mk)

ii) Give two reasons for carrying out the operation stated in a (i) above?

(1mk)

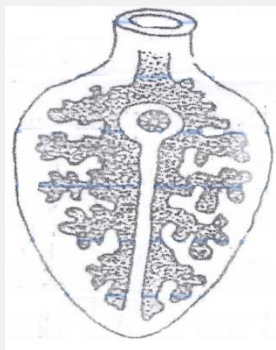
iii) At what age should the operation in a (i) above be carried out?

(1mk)

iv) Give two methods of carrying out the operation in a (i) above.

(2mks)

19. Below is an illustration of an internal parasite in livestock.



a) Identify the parasite

(1mk)

b) Name one final host for the parasite.

(1mk)

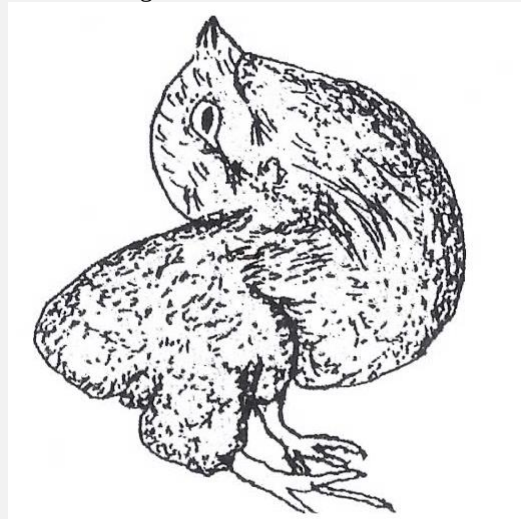
c) Name the intermediate host for the parasite.

(1mk)

d) State two measures used to control the parasite.

(2mks)

20. The following is an illustration of a chick suffering from malnutrition



a) Identify the mineral deficiency shown by the chick.

(1mk)

b) Apart from the symptom illustrated above, give three other symptoms of mineral deficiency in poultry.

(3mks)

c) Suggest a method of controlling the disorder above.

(1mk)

SECTION C

Answer two questions in this section

21. a) Describe the digestion of grass in the rumen of a ruminant. (10mks)
b) Describe five maintenance practices that should be carried out on an ox-drawn mould board plough. (5 mks)
c) State five indicators that can be observed on a goat to confirm sickness. (5mks)
22. a) Describe the procedure followed when hand spraying cattle to ensure effective use of acaricides to control ticks (10 mks)
b) What are the advantages of a four stroke cycle engine. (5mks)
c) Explain the maintenance practices of a cattle dip (5mks)
23. Explain the uses of various hand tools in the construction of a bee hive. (10 mks)
b) Give five disadvantages of free range system of rearing poultry (5mks)
c) Explain the different methods used to stimulate milk let down in a lactating cow when milking (5mks)

MODEL TEST PAPER 2

PAPER I

TIME: 2 HOURS

SECTION A (30MARKS)

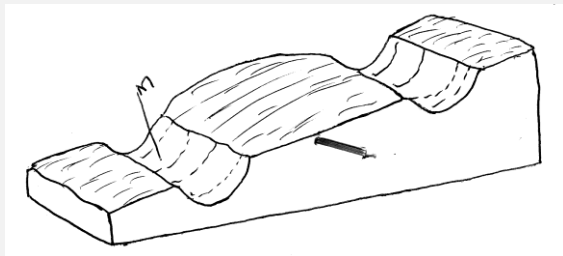
Answer ALL the question in this section in the spaces provided

1. List **four** branches of Agriculture (2mks)
2. State **three** characteristics of shifting cultivation (1 ½ mks)
3. Give **two** forms of collective land tenure system in Kenya (1mk)
4. State **four** factors to be considered before deciding on irrigation in crop production (2mks)
5. State **two** causes of hard pans in a crop field (1mk)
6. Outline **four** advantages of undersowing in pasture production (2mks)
7. Give **four** ways of controlling weeds in a field of maize (2mks)
8. Define the following terms
(a) Economic injury level (1mk)
(b) Integrated pest management (1mk)
9. Name **four** effects of applying nitrogenous fertilizers in access to crop growth (2mks)
10. State **three** ways by which overheating can be prevented in the process of making silage. (1½ mk)
11. Name the disease which causes a mass of dark spores on the flowering parts of maize? (1mk)
12. Differentiate between budding and grafting (2mks)
13. State **four** ways of improving the labour productivity of farm (2mks)
14. Give a reason for each of the following practices
(a) Top dressing established crops (1mk)
(b) Topping in pasture management (1mk)
(c) Stooking in maize production
15. State **four** factors that determine the stage at which a grain crop is ready for harvesting (2mks)
16. Give **two** fresh market varieties of carrots (1mk)
17. State four characteristics that make a crop suitable for green manuring (2mks)

SECTION B (20Mks)

Answer all questions in this section

- 18.(a) Define the term straight fertilizer (1mk)
A farmer was advised to apply 400kg of a complete compound fertilizer whose fertilizer grade is 25:20:5
(i) Calculate the amount of K₂O that was applied by the farmer (2mks)
(ii) What do figures 25:20:5 stand for (3mks)
(iii) The active ingredients in a compound fertilizer adds up to 50kgs for every 100kg of fertilizer . Account for the remaining mass (1mk)
19. The diagram below is a type of a terrace

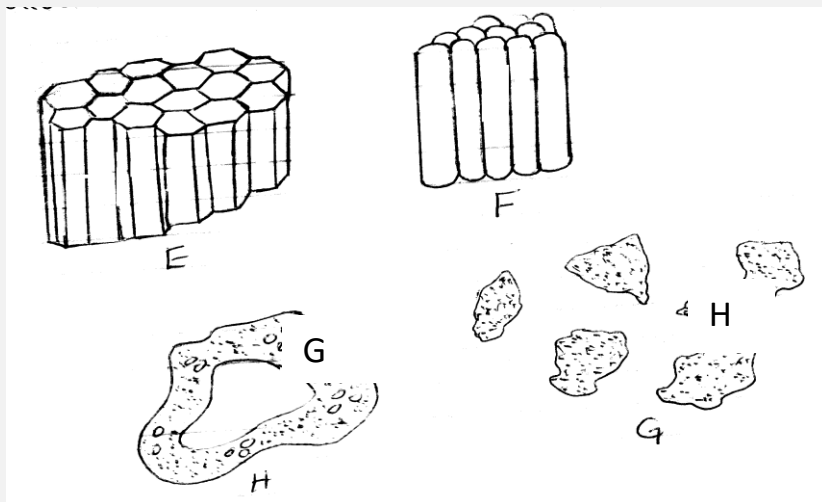


- (a) Name the type of terrace illustrated above

(1mk)

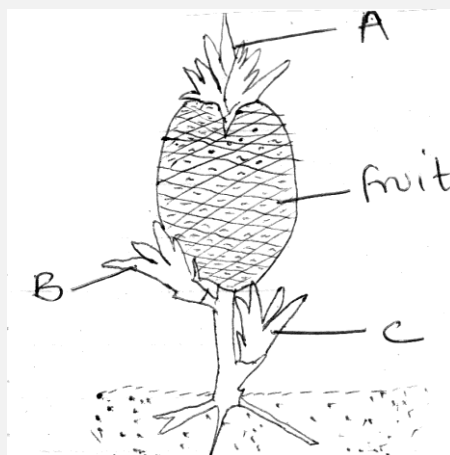
- (b) Give the name of the part labeled M in the diagram (1mk)
 (c) Apart from the above named structure state any other **four** type of terraces (4mks)

20. The diagram labeled E,F,G and H , illustrate some soil structures . Study the diagram carefully and answer the questions that follows –



- (a) Name the soil structures E,F,G and H (2mks)
 (b) State **four** field practices which can destroy the soil structures (2mks)

21. The diagram below illustrates a pineapple crop. Study it and answer the questions that follow: -



- (a) Name the parts labeled A and B (2mks)
 (b) Give **two** reasons why part C is the most preferred by farmers for propagating pineapples (1mk)

SECTION C (40 MARKS)

Answer any TWO questions in this section in the spaces provided after question 24

22. (a) Describe production of Napier grass under the following sub-heading

- (i) Varieties (2mks)
 (ii) Land preparation (4mks)
 (iii) Planting (2mks)

- (iv) Harvesting (2mks)
- (b) Explain the importance of Agroforestry trees (5mks)
- (c) State **five** factors considered when designing a crop rotation programme (5mks)
- 23.(a) A farmer has 160ha of land, 60ha of which is under wheat, 32 ha under maize, 12ha under fodder crops and the rest under pastures. The farmer wishes to know whether replacing 12ha of maize with irish potatoes the following year would be worthwhile. The fertilizer would have to be increased from 5 bags per hectare per hectare for maize to 7 bags per hectare for irish potatoes and an extra 100 man – days of casual labour per hectare will be necessary as a result of the change. The average yield of maize and irish potatoes is 45 and 115 bags per hectare respectively. The prices are Ksh. 1,400 per bag for maize and Ksh . 1,200 per bag for irish potatoes, seed costs are Ksh. 2,500 per hectare for maize and Kshs 30,000 per hectare for irish potatoes fertilizer costs are Kshs 1,300 per bag . Labour is paid to Ksh. 150 per man-day/ha . Prepare a partial budget and indicate whether the change is worthwhile (12mks)
- (b) State **eight** factors of Land Fragmentation in Kenya (8mks)
24. (a) Discuss the various structural measures used to conserve soil and water in the farm (10mks)
- (b) Explain **five** reasons for carrying out minimum tillage (5mks)
- (c) Outline **five** benefits of tissue culture in crop propagation (5mks)

PAPER 2

SECTION A (30 MARKS)

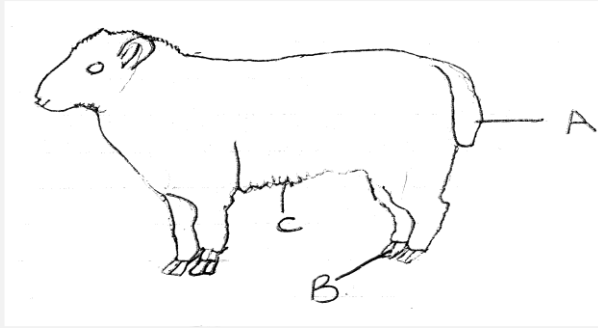
Answer ALL questions in this section in the space provided

1. Give **three** maintenance practices carried out on crosscut saw (1 ½ mks)
2. Mention **four** characteristics of exotic breeds of cattle (2mks)
3. Name **one** mineral deficiency as a result of each of the following symptoms
 - (a) Soft shelled eggs (½ mk)
 - (b) Pica (½ mk)
 - (c) Grass tetany /staggers (½ mk)
 - (d) Paraketosis (½ mk)
4. Name the infective stage of each of the following internal parasite
 - (a) Liverfluke (½ mk)
 - (b) Tapeworm..... (½ mk)
5. Differentiate between a roughage and a concentrate in livestock nutrition (2mks)
6. State **four** reasons for steaming up in dairy cattle management (2mks)
7. State **three** male characteristics that a cock loses after caponization (1½mks)
8. Outline three advantages of using concrete in construction of farm structures (1½mks)
9. What do you understand by the term incubation period as used in livestock diseases (1mk)
10. (i) Mention **four** ways in which stress can be avoided in a flock of layers (2mks)
11. Give **four** sources of farm power that are environmentally friendly (2mks)
12. Name **three** marketing channels that market Beef products in Kenya (1 ½ mk)
13. State **two** morphological differences between Bactrian and dromedary camel breed (2mks)
14. Give the function of the following parts of a petrol engine system
 - (a) Connecting rod (½ mk)
 - (b) Carburetor (½ mk)
 - (c) Thermostat (½ mk)
 - (d) Piston rings (½ mk)
 - (e) Radiator (½ mk)
15. Explain how each of the following factors can be applied in the control of livestock disease
 - (a) Quarantine (1mk)
 - (b) Proper selection and breeding (1mk)
16. Outline **four** effects of liverflukes on sheep (2mks)
17. Highlight **three** disadvantages of inbreeding (1 ½ mks)

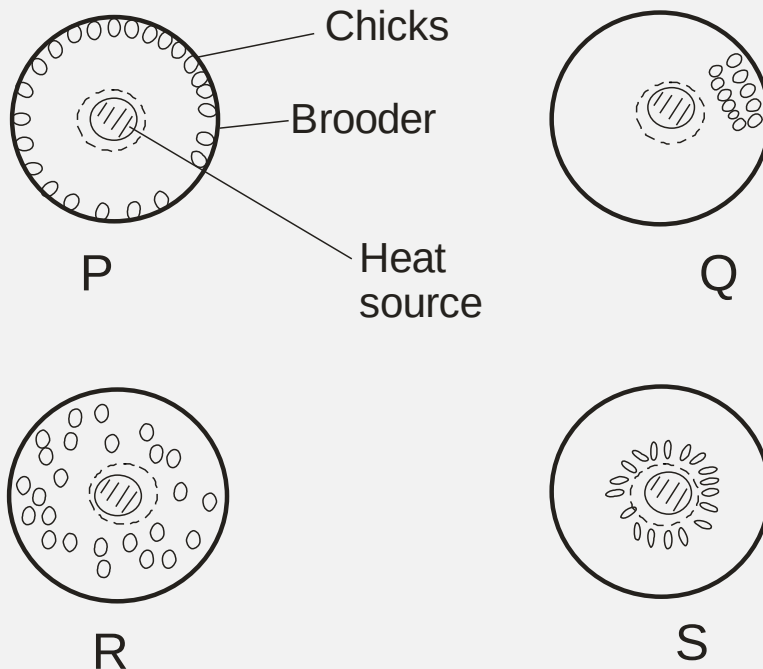
SECTION B (20mks)

Answer all the questions in this section

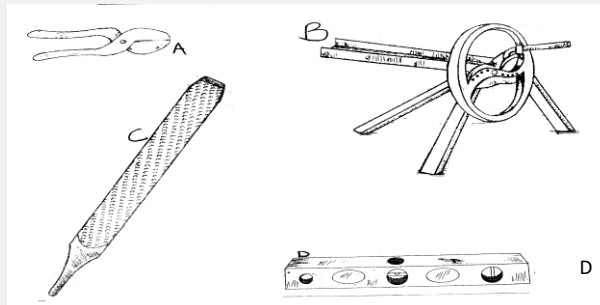
18. A farmer wants to prepare 500kg of calf rearing ration containing 22% DCP using rice bran (10%DCP) and simsim seedcake (35% DCP) .How much of each feedstuff should the farmer purchase to prepare the ration (show your working) (5mks)
19. The diagram below illustrates livestock rearing practices. Study it carefully and answer the questions that follow:-



- (a) Identify the operation carried out on the part labeled A (1 mk)
 (b) Give **two** reasons for carrying out the operation identified in (a) above (2mks)
 (c) Mention **one** problem that would occur if the operation that should be carried out on part labeled B is not carried out (1mk)
 (d) State **two** precautions that should be observed when shearing wool on part labeled C (2mks)
20. The diagrams below illustrate behavior of chicks in a brooder



- (a) Describe the behavior of chicks in brooders labeled Q, R and S (3mks)
 (b) Mention **two** other observable behaviour in chicks in brooder labeled P other than the behavior shown above
21. The illustrations below represent farm tools/equipment. Study them carefully then answer the questions that follow



- (a) Identify tools/equipment labeled C and D (2mks)
 (b) State the use of tool A (1mk)
 (c) In which category of farm tools and equipment does not tool belong? (1mk)

SECTION C (40 MARKS)

Answer any two questions in this section in the spaces provided after question 24

22. (a) Describe **six** operational differences between mould board plough and a disc plough (6mks)
(b) Outline the procedure for harvesting honey from Kenya Top bar line (10mks)
(c) State **four** ways of stimulating a cow for milklet down (4mks)
23. (a) Discuss bloat disease under the following subheadings
(i) Causes (3mks)
(ii) Symptoms (3mks)
(iii) Control Measures (4mks)
(c) Describe **five** signs of good health in livestock (10mks)
- 24 (a) Describe the feeding practices carried out on a calf from birth to weaning (10mk)
(b) Mention **four** factors affecting maintenance requirements in livestock nutrition (4mks)
c) Describing **six** factors to consider when siting farm structures (6mks)

MODEL TEST PAPER 3
AGRICULTURE

Paper 1

Kenya Certificate of Secondary Education

SECTION A (30 marks)

1. State four ways by which plant nutrients are lost. (2 marks)
2. Explain the meaning of the term production function. (1 mark)
3. Name four product relationships in Agriculture. (2 marks)
4. State four uses of a green house in a farm. (2 marks)
5. Name four farming practices farmers can use to achieve minimum tillage. (2 marks)
6. Outline four importance of mulching in Agricultural land. (2 marks)
7. Name four sources of Agricultural credit. (2 marks)
8. Give four ways of harvesting water in the farm. (2 marks)
9. Name the form in which the following nutrients are absorbed by plants. (2 marks)
 - i. Nitrogen
 - ii. Phosphorus
 - iii. Sulphur
 - iv. Calcium
10. State four sites Agroforestry trees can be planted in the farm. (2 marks)
11. Outline four factors affecting the efficiency of pesticides. (2 marks)
12. Name a crop that require the following training.
 - a. Propping (½ marks)
 - b. Threlishing (½ marks)
 - c. Staking (½ marks)
13. List four sources of organic matter in the soil. (2 marks)
14. Give advantages of tissue culture. (2 marks)
15. Outline four factors that influence the supply of casual labour on the farm. (2 marks)
16. State three properties of soil that are influenced by its texture. (1½ marks)

SECTION B

Answer all questions in the spaces provided.

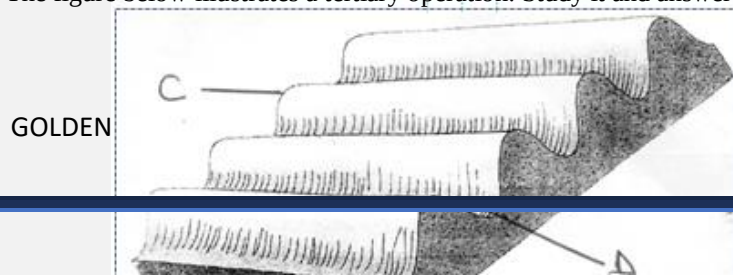
17. Below are weeds. Study them and carefully answer the questions that follow.



Identify weeds labeled S,T,U and P.

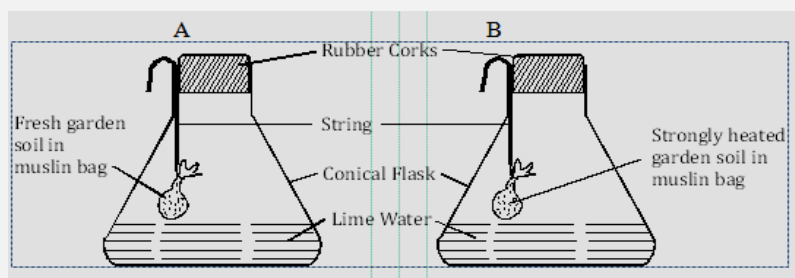
(4 marks)

18. The figure below illustrates a tertiary operation. Study it and answer the question that follow.



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- a. Identify the tertiary operation illustrated above. (1 mark)
- b. Name the part labeled C and D. (2 marks)
- c. Name two other tertiary operations that can be carried out in the farm apart from the one shown above. (2 marks)
19. A form one student set up experiment as shown below to study an aspect of soil. The set up was left for six hours. Study it and answer the questions that follow.



- a. What was the aim of the experiment? (1 mark)
- b. State two observations that were made in each of the flasks C and D. (2 marks)
- c. Why was the soil in flask D strongly heated? (1 mark)
- d. Outline ways in which soil living organisms affect agricultural production. (2 marks)
20. The table below shows the population and gross domestic products of countries A and B.

Country	Gross Domestic product (millions)	Population (millions)
A	1800	36
B	1200	15

- a. Calculate the Per Capita income for each country. Show your working. (2 marks)
- b. Which of the following two countries is more developed economically. (½ mark)
- c. How can Agriculture increase the gross Domestic product of a country. (½ mark)

SECTION C 40 marks)

Answer any two questions on this section in the spaces provided after the last question.

21. Describe production of dry beans under the following subheading. (5 marks)
- Land preparation. (7 marks)
 - Planting (3 marks)
 - Harvesting (5 marks)
 - Outline any five post harvesting practices carried out on beans after harvesting. (4 marks)
22. a. Outline various ways through which vegetation cover reduces soil erosion. (4 marks)
- b. Outline various conditions which necessitate land clearing. (4 marks)
- c. Outline various advantages of land consolidation. (6 marks)
- d. Give six benefits of planting annual crops early. (6 marks)
23. The following table shows production of maize using various levels of input.

Units of variable input (man days)	Total output	Marginal product	Average product
0	0	-	-
1	6	-	-
2	18	-	-
3	33	-	-
4	40	-	-
5	45	-	-
6	48	-	-
7	48	-	-
8	40	-	-

- Work out the marginal product and the average product and fill in the table. (9 marks)
- On a graph paper, plot graphs showing total output marginal product and average product against the variable input. (3 marks)
- On the graph, draw lines to show the following zones of production.
 - Increasing returns production function.
 - Decreasing returns production function.
 - Diminishing return zone.

PAPER 2

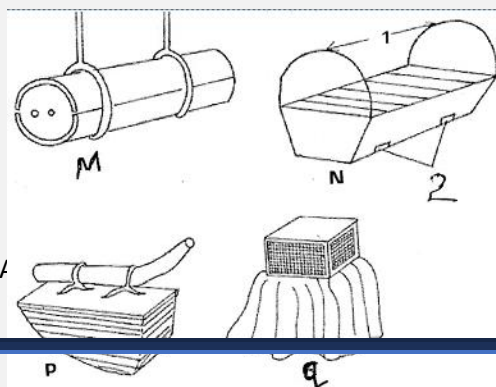
Answer ALL questions in this section on the spaces provided.

- Name the tool or equipment that is complementary in use with : (2 marks)
 - Canula
 - Elastrator
 - Lead stick
 - Cold chisel
- List four factors considered when formulating livestock rations. (2 marks)
- State six methods a farmer can use to control livestock diseases. (3 marks)
- State four advantages of using corrugated iron sheets for roofing instead of asbestos (2 marks)
- State four practices through which a farmer would ensure that no engine knock occurs in a tractor. (2 marks)
- State four practices that a farmer should carry out to reduce egg eating in poultry house. (2 marks)
- State four advantages of carrying out raddling in sheep management. (2 marks)
- Outline four importance of additives in livestock feeds. (2 marks)
- Give four control measures of milk fever. (2 marks)
- Give four characteristics of a heavy poultry breeds. (2 marks)
- Give four reasons why tilapia fish is commonly reared in Kenya farm ponds. (2 marks)
- Give four reasons for seasoning timber before use. (2 marks)
- Give four factors which affects digestibility. (2 marks)
- Outline four control measures of liver fluke. (2 marks)
- Differentiate between mothering ability and prolificacy. (1 mark)

SECTION B (20 MARKS)

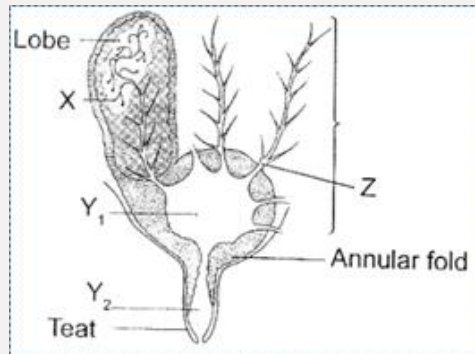
Answer all the questions in this section in the spaces provided.

- Diagram M, N, P and Q shows some structures used in a apiculture. Use them to answer the questions that follow.



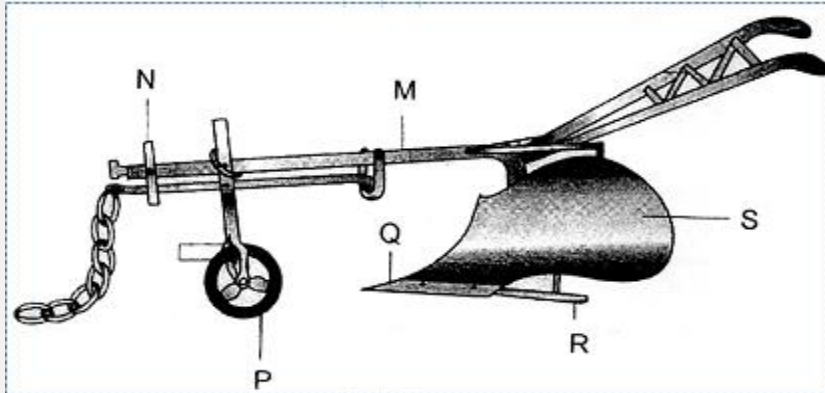
- a. Identify the structure labeled M, N, P, Q. (2 marks)
- b. State the use of equipment P and Q. (2 marks)
- c. State one advantage structure N has over structure M. (1 mark)
- d. State the use of the parts labeled 1 and 2 on structure N. (2 marks)

17. The diagram below illustrates a cross section of a cow's udder. Study it carefully and answer the questions that follow.

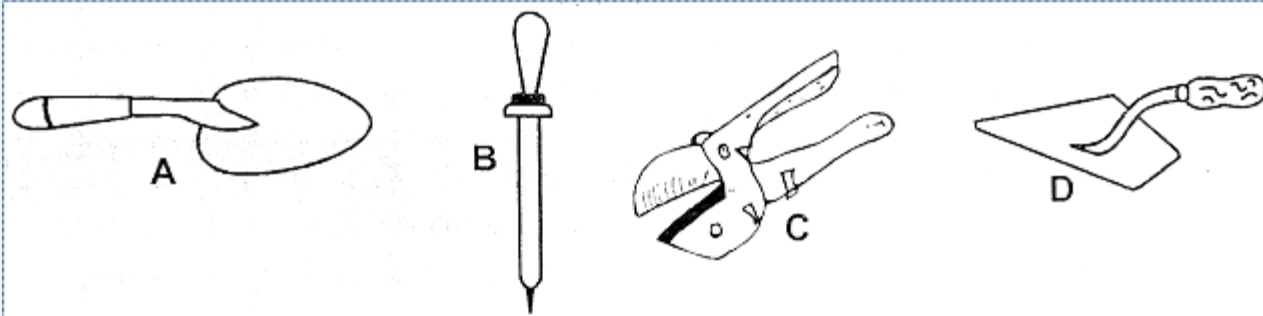


- a. Identify the parts labeled: (2 marks)
 - X
 - Y₁
 - Y₂
 - Z
- b. Name two hormones that control milk let down in a cow. (1 mark)

18. Below is an illustration of a farm implements. Study it carefully and answer the questions that follow.



- a. Identify the farm implements. (1 mark)
 - b. Name the parts labeled N, P, Q, R and S. (5 marks)
 - c. Give the function labeled P and S. (2 marks)
19. Study the diagrams below and answer the questions that follows.



Identify tools A, B, C and D.

(2 marks)

SECTION C (40 MARKS)

Answer two questions in this section on the spaces provided after question 22.

20. a. Explain the operational differences between disc plough and mould board plough. (12 marks)
- b. Outline 8 microbial activities that take place in the rumen (8 marks)
21. a. Describe the characteristics which eggs for incubation should possess. (6 marks)
- b. Outline various methods used to control cannibalism in poultry management. (8 marks)
- c. Outline 6 advantages of artificial calf rearing. (6 marks)
22. i. Discuss fish farming in a fish pond under the following sub headings.
 - a. Stocking the pond (4 marks)
 - b. Feeding the fish (4 marks)
 - c. Harvesting fish (4 marks)
 - d. Maintenance of the pond. (4 marks)
- ii. Outline precaution observed when using tools and equipment. (4 marks)

MODEL TEST PAPER 4

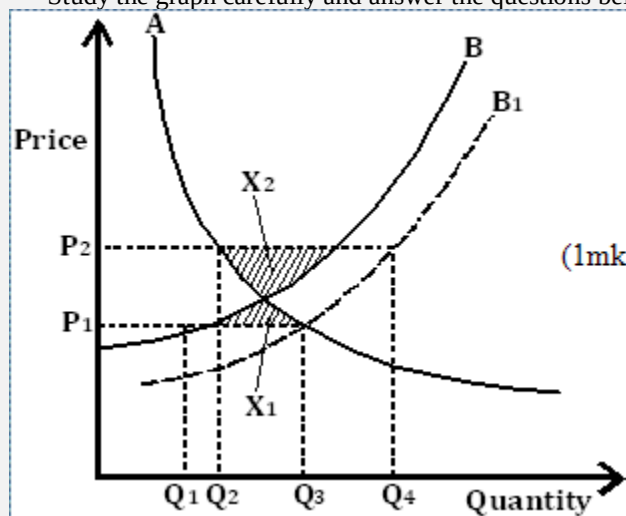
PAPER 1

1. State four soil factors that may influence the crop grown by a farmer. (2mks)
2. State four problems that would be experienced by a farmer for persistent use of herbicides in crop production. (2mks)
3. Outline four environmental factors which favour the population of pests in a given locality. (2mks)
- 4 a) What is land reform? ($\frac{1}{2}$ mk)
- b) Name three practices of land reform in Kenya today. (1½mks)
5. a) What is elasticity of demand? ($\frac{1}{2}$ mk)
- b) Give three reasons why an agricultural commodity may be price in-elastic. (1½ mks)
6. a. Differentiate the following.
 - i) Topping from Top dressing in pasture management. (1 mk)
 - ii) Itinerant Traders from broker agents (1 mk)
 - iii) Weir from a Dam in soil and water management. (1 mk)
 - iv) Agro-silviculture from Agro-silvopastoral. (1 mk)
7. Give two reasons for carrying out rogueing in a crop field. (1 mk)
8. a. What is a soil sample? ($\frac{1}{2}$ mk)
- b) Outline three activities that are carried out during soil sampling. (1½mks)
9. List down two maintenance practices of sub-surface irrigation. (1mk)
- 10 a) State the law of equimarginal returns in a production process. ($\frac{1}{2}$ mk)
- b) State three un-desirable effects of weeds in pastures. (1½mks)
11. Given that tomatoes are planted at a spacing of 60cm by 30cm. Calculate the plant population in a plot of land measuring 12m by 9m. Show your working. (2mks)
- 12 a) what is labour peak in the farm? ($\frac{1}{2}$ mk)
- b) State three ways in which labour peaks can be overcome in the farm. (1½mk)
- 13 a) what is a land title deed? ($\frac{1}{2}$ mk)
- b) State four advantages of a land title deed to a farmer. (2mks)
14. Calculate the amount of P_2O_5 contained in 200kg of a compound fertilizer 25:10:5 (2 mks)
- 15 a) what is a farm budget? ($\frac{1}{2}$ mk)
- b) State three benefits of budgeting to a farm manager. (1½ mks)

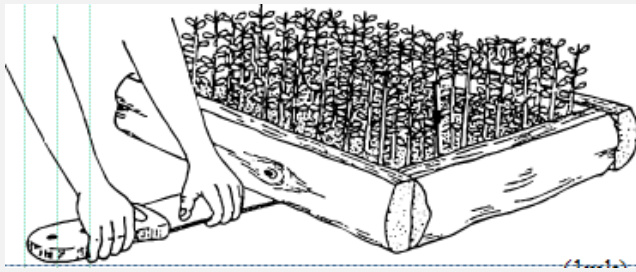
SECTION B (20 MARKS)

Answer All the questions in this section

16. Alongside is a graph showing the relationship between two market forces represented by curves A and B in price determination. Study the graph carefully and answer the questions below.



- a) i) Name the curves labelled A and B. (1mk)
 - ii) Give the name, and state the significance of the point where curves A and B intersect. (1mk)
 - iii) Outline the significance of the areas shaded X_1 and X_2 . (1mk)
 - b) Outline two factors that could shift curve B to B_1 in a perfect market. (1mk)
17. The diagram shows a nursery management practice carried out on seedlings. Study it carefully and answer the questions that follow:-



a) Identify the management practice.

(1mk)

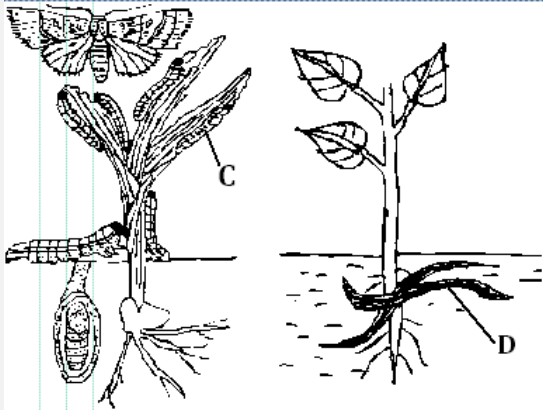
b (i) Why is it necessary to carry out the above practice? Give two reasons.

(2mks)

ii) State one problem associated with the above practice.

(1mk)

18. The illustrations represent two plants attacked by pests. Study them carefully and answer the questions that follow:-



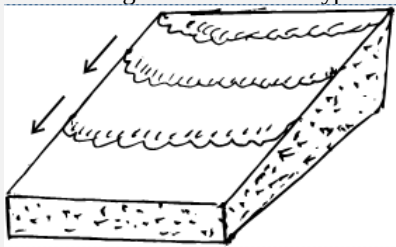
a) i) Identify pests C and D

(1 mark)

ii) Outline the most likely damage that can be caused by pests C and D.

(1mk)

19. The diagram illustrates a type of erosion. Carefully study it, and answer the questions that follow:-



a) Identify the type of erosion.

(½mk)

b) i) Name the best two structures used in controlling the above type of erosion.

(2mks)

ii) Outline three effects of the above type of erosion in the farm.

(1½mks)

20. The diagram below represents weeds. Study them carefully and answer the questions that follow.



- a) Identify the weeds labelled E and F. (1mk)
- b) i) Give two reasons why weed E is more harmful in a field of wheat than weed F. (2mks)
- ii) State the significance of weed F in livestock production. (1mk)

SECTION C (40 MARKS)

Answer any TWO questions from Section C

- 21 a) Outline the principles governing co-operative societies in Kenya. (5mks)
- b) Name and explain methods of fertilizer application. (5mks)
- c) Discuss the common micro-catchments used in soil and water conservation in tree crops. (5mks)
- d) Explain the importance of a balance sheet in a farming business. (5mks)
22. Describe the production of tomatoes under the following headings.
- a) Nursery preparation. (12 mks)
- b) Transplanting (5mks)
- c) Field management practices. (3mks)

23. Maboga farm started the year on 1: 2011 and as at 31. 12. 2011 had the following transactions.

	Kshs.
Sales of poultry	5,000.00
Opening valuations	10,000.00
Purchase of seeds	2,800.00
Purchase of fertilizers	2,000.00
Sales of Beans	1,200.00
Sales of vegetables	500.00
Closing valuations	15,000.00
Casual laborers (wages)	2,500.00
Depreciation (Tractors)	500.00
Interest (Borrowed loan)	400.00
Purchase of day old chicks	800.00
Sales of milk	6,000.00
Purchase of calves	10,000.00
Miscellaneous Expenses	1,000.00

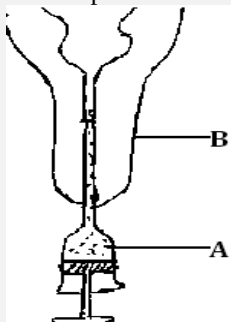
- a) Prepare a profit and loss Account for Maboga farm. (10mks)
- b) Did the farm qualify for credit? Give one reasons. (2mks)
- c) Describe the procedure followed during ensiling. (8mks)

PAPER 2

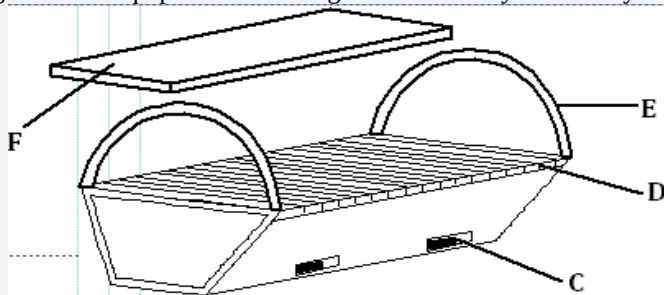
1. Give the name used to describe the following animals. (1½mks)
i) Pig that has given birth several times.
ii) Castrated male bird (domestic fowl)
iii) A young one of a rabbit after birth.
2. Name the breed of a dairy cow with light yellow colour, to a shade of black, protruding black eyes with a black muzzle and tail switch. (1mk)
3. Name the intermediate host of the following. (1mk)
i) Taenia saginata .
ii) Fasciola gigantica .
4. a) State one use of each of the following farm tools. (1mk)
i) Mallet
ii) Elastrator
- b) State the functional difference between a marking gauge and mortise gauge. (1mk)
- c) State four problems associated with a farmer who does not maintain his farm tools and equipments. (2mks)
5. Outline four duties of a water bee in a bee colony. (2mks)
6. State four causes of infertility in dairy cattle. (2mks)
7. Outline four factors that determine the nutritional requirement of cattle. (2mks)
8. State four practices that can be carried out to control egg breaking and drinking by birds in a deep litter system. (2mks)
9. Why is it necessary to weigh piglets immediately after farrowing and weaning? Give four reasons. (2mks)
10. List four specific symptoms of east coast fever diseases in cattle. (2mks)
11. State four disadvantages of wind power. (2mks)
12. Name two methods of timber treatment. (2mks)
13. State four disadvantages of using thatch as roofing material. (2mks)
14. a) what is depreciation in farm tools & equipments. (½mk)
b) State two possible causes of depreciation in farm tools and equipment. (1mk)
15. Name four components of the ignition system in a tractor engine. (2mks)
16. Why is it necessary to feed young mammals with colostrum for the first two weeks after birth? Give three reasons. (1½mks)

SECTION B (20 MARKS)

17. The diagram below represents a routine practice carried out on dairy animals.



- a) Identify the practice illustrated above. (½mk)
b) i) Name the material labelled A in the illustration above. (1mk)
ii) Give three reasons why it's necessary to apply milking jelly in the part labelled B. (1½mks)
18. Below is a diagram of an equipment used in agriculture. Study it carefully and answer the questions that follow.

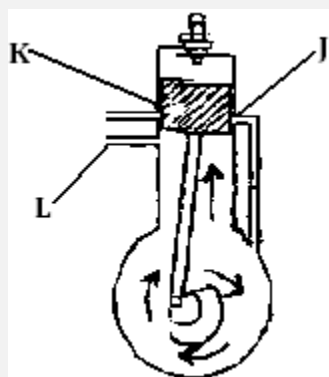


- a) Identify the equipment. (1mk)
b) Name the parts labelled C, D E. (1½mks)

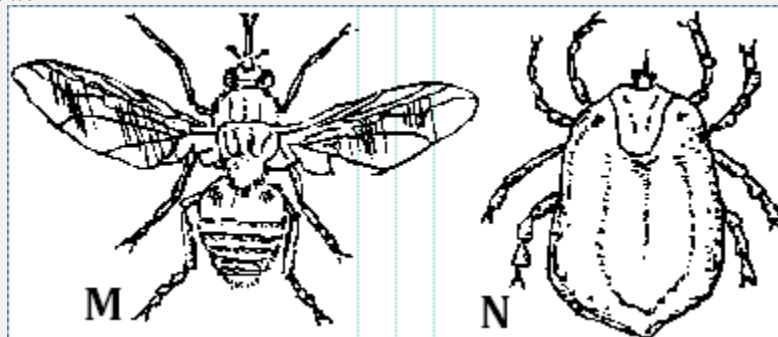
- c) i) State one role played by the parts labelled C, D, E and F (4mks)
 ii) Name other two types of equipments which serve the same function as the one illustrated above. (1mk)
19. The diagram below represents a practice carried on birds.



- a) State the purpose of the practice. (½mk)
 b) Identify with a reason which of the two birds is desirable. (1mk)
20. The diagram below represent a stroke in a two stroke cycle Engine. Study it carefully and answer the questions that follow:

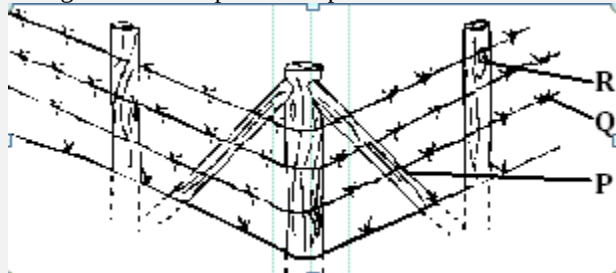


- a) i) Identify the stroke. (½mk)
 ii) Name the parts labelled J, K and L. (1½mks)
- b) State what is happening in the parts labelled K and L during this stroke. (1mk)
21. The diagrams below represent common parasites in livestock production. Study them carefully and answer the questions that follow.



- a) Identify the parasites labelled M and N. (1mk)
- b) i) Name one disease transmitted by parasites M and N in livestock production. (1mk)
 ii) Outline one non-chemical methods of controlling parasites M and N. (1mk)

22. The diagram below represents a part of fence in a farm. Study it carefully and answer the questions that follow.



- a) Identify the part of the fence represented. (½mk)
- b) i) Name the parts labelled P and Q. (1mk)
- ii) State three disadvantages of the type of the fence represented above. (1½mks)
- SECTION C (40 MARKS)**
- 23 a) Compare the use of an-oxen drawn plough with a tractor drawn mouldboard plough. (10 mks)
- b) Outline the procedure followed in establishing the foundation of a permanent farm building. (10 mks)
24. Describe the rearing of layers in a deep litter system from the point of lay. (20mks)
- 25 a) i) Describe the life cycle of tapeworms. (8mks)
- ii) State and describe two measures of controlling tapeworms. (2mks)
- b) State and describe the general measures undertaken to control livestock diseases. (10mks)

MODEL TEST PAPER 5

PAPER 1

1. Give two reasons why farmers are encouraged to practice organic farming. 1mk
- 2) State two advantages of earthing up in crop production. 1mk
- 3) Give four disadvantages of landlordism and tenancy system of land tenure. 2mks
- 4) How are the following conditions prevented in crop production? 2mks
 - i) Forking
 - ii) Greening of root tuber.
- 5) Give two reasons why farmers keep farm accounts. 1mk
- 6) Give a reason for storing water for 36hrs during water treatment. ½ mk
- 7) List four natural factors that encourage soil erosion. 2mks
- 8) Define the following terms as used in pasture establishment. 1mk
 - i) Nurse crop
 - ii) Oversowing
- 9) Give a reason for each of the following during compost making . 2mks
 - i) Adding soil
 - ii) Adding farmyard manure
 - iii) Adding ash
 - iv) Adding calcium ammonium nitrate.
- 10) State two biotic factors that are useful in agricultural production. 1mk
- 11) List four post harvest practices that are carried out in maize production. 2mks
- 12) State three functions of Nitrogen in crops 1 ½ mks
- 13) State four advantages of overhead irrigation. 2mks
- 14) Outline two factors that could determine the amount of fertilizers to be top-dressed to a crop in the field. 1mk
- 15) State four qualities of a parent plant which should be considered when selecting vegetative material for propagation. 2mks
- 16) Distinguish between Gross Domestic Product (GDP) and Gross National product (GNP). 2mks
- 17) List three benefits of a tithe deed to a farmer. 1 ½ mks
- 18) State two requirements of a successful tissue culturing in crop production. 2mks
- 19) Give three methods of practicing rotational grazing. 1 ½ mks

SECTION B (20 Mks) COMPULSARY

- 20) The diagram below shows a common weed. study it and answer the questions that follow.

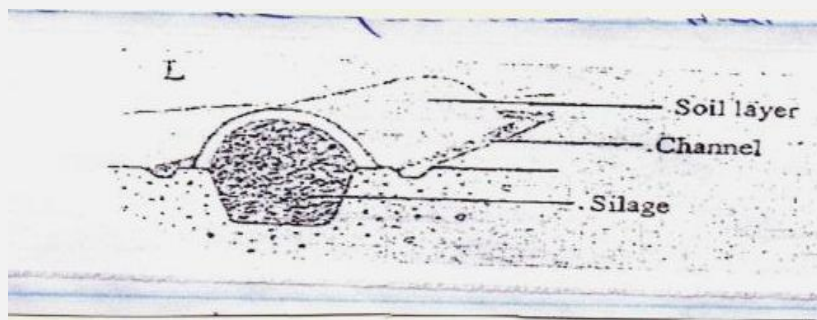


- a) Identify the weed illustrated above . 1mk
- b) Give a reason as to why it is difficult to control the weed illustrated above . 1mk

- 21) The farm record below was extracted from a farmer's book. Study it and answer the questions that follow.

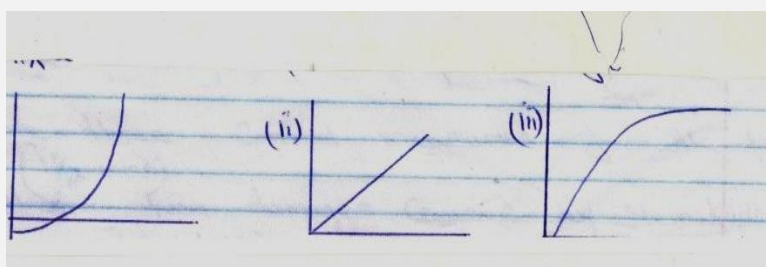
Name of person	Payroll No.	DAYS	Days worked	Rate of pay Ksh	Total pay(Ksh)	Sign of workers
		1 2 3 4				
DAVID	01		20	@ 100	2000/=	
JULIE	02		25	@ 200	2500/=	

- 22) Observe the structure labeled L and answer the questions that follow.



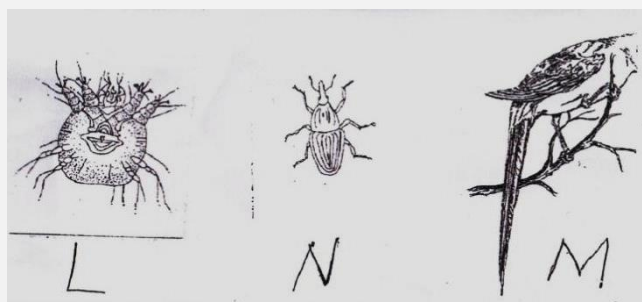
- a) Name the structure labelled L _ 1mk
- b) State the importance of the channel around the structure L. 1mk
- c) Outline two reasons for conserving forage . 2mks
- d) Name two main silage losses that occur ensiling. 1mk

23) Below are graphical representation showing different types of production functions. Study them and answer the questions that follow.



- a) Identify the types of production function represented by the diagrams. 3mks
- b) What is meant by production function as used in agricultural economics? 1mk

24) The diagrams below illustrate both field and storage pests.



- a) Identify the pests in the illustration. 1 ½ mks
- b) State ways in which pests L and M cause damage to crop. 1mk
- c) State three control measures for the pest N 1 ½ mks
- d) Outline two damages caused by Nematodes on crops. 1mk

SECTION C (40)

ANSWER ANY TWO QUESTIONS IN THIS SECTION

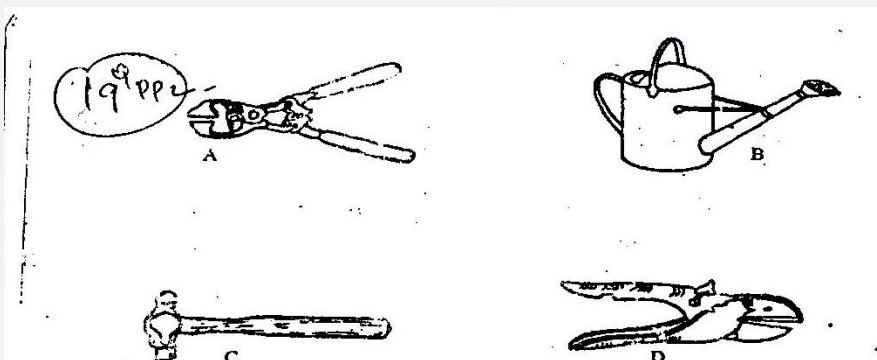
- 25) a) Discuss maize production under the following sub-headings:-
 - i) Ecological conditions. 3mks
 - ii) Field preparation. 3mks
 - iii) Planting and field management. 4mks
- b) Describe transplanting of tomatoes. 10mks
- 26) a) Describe five management practices carried out on napier grass to maximize production. 5mks
- b) Explain five ways in which draining marshy/land encourage crop growth. 5mks
- c) Describe briefly the effects of mass wasting/solifluction
- 27) a) Describe the procedure of soil sampling. 10mks
- b) Explain the reasons for establishing settlement schemes in Kenya soon after independent. 5mks
- c) Explain five ways through which soil loses fertility. 5mks

PAPER 2

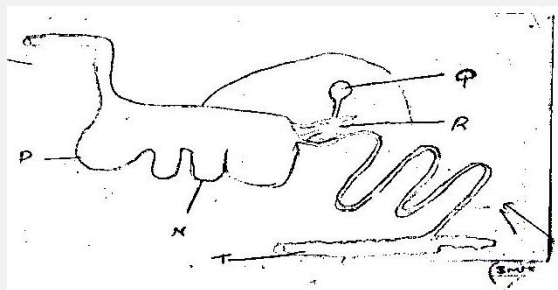
- 1) Name a beef with the following characteristics. 1mk
 - a) Black in colour
 - b) Is polled
 - c) Has a long cylindrical compact body.
- 2) Give five reasons for identification in cattle management. 2 ½ mks
- 3) State four qualities of eggs preferred by consumers in the market. 2mks
- 4) List four predisposing factors of livestock diseases. 2mks
- 5) Differentiate between isolation and quarantine in livestock health. 2mks
- 6) Name three points of the three point hitch implements. 1 ½ mks
- 7) State one role of damp proof course in the foundation of a farm building. ½ mk
- 8) List five signs of farrowing in a sow. 2 ½ mks
- 9) Differentiate between milk secretion and milk let down. 2mks
- 10) State four reasons why lubrications is important in a tractor system. 2mks
- 11) Name two livestock diseases that may be controlled by use of artificial insemination. 2mks
- 12) Name four uses of camels. 2mks
- 13) Name one intermediate host of each of the following endoparasites. 1mk
 - a) Roundworms
 - b) Liver-flukes
- 14) List five practices carried out on wooden fences to make them last longer. 2 ½ mks
- 15) Name two feed additives that can be included in livestock diet. 1mk
- 16) Give one use of robot plane in the workshop. 1mk
- 17) List down any four financial documents kept by a farmer. 2mks
- 18) Name two types of inventory records used in farm records. 2mks

SECTION B

- 19) Below are illustrations of farm tools and equipments.

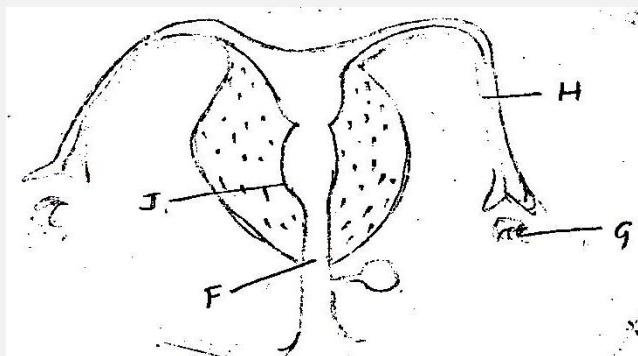


- a) Identify the tool/equipment labeled (A and B) 1mk.
 - b) State two appropriate uses of the tools labelled C. 1mk
 - c) Mention the maintenance practices that should be carried out on D. 1mk
- 20) The diagram below shows the digestive system of cattle. Study it and answer the questions that follow.



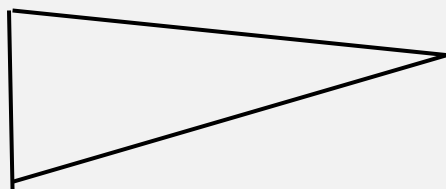
- a) Name the parts labeled N, P and Q. 3mks
- b) State one function for each of the parts labeled S and T. 2mks
- c) Give one enzyme produced by each of the parts labeled R and S. 2mks

21) The diagram below show the reproduction system of a cow. Study it carefully and answer the questions that follow.



- | | |
|---|------|
| a) Name the part labeled F and H. | 2mks |
| b) Give two functions of the parts labeled G. | 2mks |
| c) Give the role of part labeled J. | 1mk |

22) The diagram below illustrates the general shape of a cattle breed. Study it carefully and answer the questions that follow.



- | | |
|--|------|
| a) Identify the type of breed illustrated by the above shape. | 1mk |
| b) Give examples of the breed identified in (a) above. | 2mks |
| c) State four physical characteristics of the type of breed identified in (a) above. | 2mks |

SECTION C(40 MKS)

ANSWER ANY TWO QUESTIONS FROM THIS SECTION

- | | | | |
|----|------|--|-------|
| 23 | a) | Describe the management of a sow one week before farrowing and three days after farrowing. | 12mks |
| | b) | Explain eight ways in which ticks can be controlled on a livestock farm | 8mks |
| 24 | a) | Describe the use of various hand tools required for the construction of a wooden rabbit hutch. | 10mks |
| | b) | Describe management practices that ensure clean milk production in a dairy farm. | 10mks |
| 25 | (a) | Discuss the disease foot rot under the following sub-headings:- | |
| | i) | Predisposing factors. | 2mks |
| | ii) | Symptoms of attack. | 5mks |
| | iii) | Control measures. | 3mks |
| | b) | Explain the factors considered when culling livestock. | 5mks |
| | c) | Explain the different methods of pregnancy diagnosis in cattle. | 5mks |

MODEL TEST PAPER 6

PAPER 1

SECTION A-30 MARKS

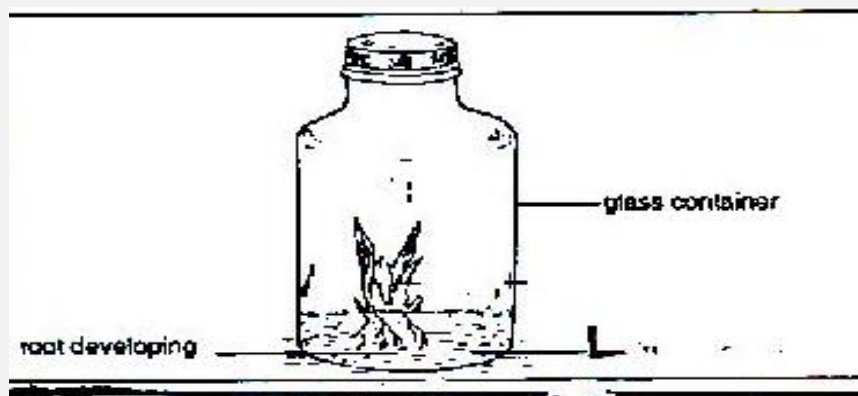
ANSWER ALL THE QUESTIONS

1. List two advantages of practicing mixed farming. (1 mark)
2. Differentiate between **aquaculture** and **apiculture**. (1 mark)
3. State four advantages of organic farming in agricultural production. (2 marks)
4. Name four aspects of rainfall that a farmer should consider during selection of a crop to grow in a given area. (2 marks)
5. State four ways through which the amount of light harnessed by a plant can be increased. (2marks)
6. State two disadvantages of superfluous water in the soil during crop production. (1 mark)
7. Give two ways through which cultural beliefs and practices affect agriculture production (2 marks)
8. Mention two ways through which the pH of an alkaline soil can be lowered. (1 mark)
9. State three conditions of land that necessitate land cleaning before primary cultivation (1½marks)
10. List three factors that determine the number of secondary cultivation done. (1 ½marks)
11. Name two methods of soil sampling. (1 mark)
12. State two of advantages of using crowns instead of suckers in pineapple production (1 mark)
13. List down four importance of budding in crop production. (2 marks)
14. State four cultural methods of water and soil conservation. (2 marks)
15. Outline four harmful effect of weeds in livestock forage production. (2 marks)
16. Give four reasons for land redistribution in Kenya. (2 marks)
17. State four ways in which labour productivity can be improved. (2 marks)
18. Outline three qualities of a good farm manager. (1½ marks)
19. State three functions of National Cereals and Produce Board(NCPB) in Kenya. (1 ½ marks)

SECTION B- 20 MARKS.

(ANSWER ALL QUESTIONS)

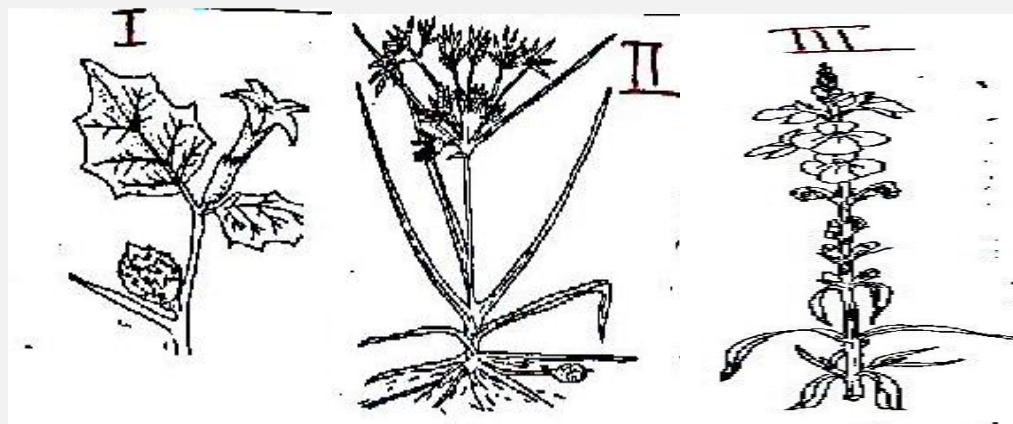
20.The diagram below shows a nursery practice.



- a.) Name the practice shown. (1 mark)
 - b.) State three advantages of the practice over any other nursery practice. (3 marks)
 - c.) State the function of L. (1 mark)
- 21.) The diagrams below represent field practices carried out in tea.
- Diagram A shows the steps of training a tea tree. Step 1: A young tea tree. Step 2: The tree is trained to a horizontal wire. Step 3: The tree is trained to a vertical wire. A label 'FINAL' is next to the final trained tree.

Diagram B shows the steps of training a tea tree. Step 1: A young tea tree. Step 2: The tree is trained to a horizontal wire. Step 3: The tree is trained to a vertical wire. Step 4: The tree is trained to a horizontal wire. A label 'FINAL' is next to the final trained tree.
- a) Name the method being undertaken in A and B. (1 mark)
 - b) State two advantages of method B over method A. (2 marks)
 - c) Name two other ways used to achieve same results as in practice B. (2 marks)

22.) The **diagrams** below **represents** various weeds that grow in crop fields. Use the diagrams to answer the questions that follow.



- a) **Name** the weed I & II (1 mark)
 b) **Give one** harmful effect of each of the weeds I, II and III (3 marks)

23.) Mr. MWANGI has **4 hectares** of arable land. **1.5 hectares** are under wheat production, **0.5 hectares** under maize, **0.3 hectare** under fodder crops and the rest is under improved grass ley. He wishes to know if replacing **0.3 hectares** of maize with potatoes in the following season would be worthwhile. Fertilizer rate would have to be increased from **2 bags per hectare to 2.5 bags per hectare for potatoes** and an extra **40 mandays** of casual labour per hectare would be required for the change. Average yield of maize and potatoes are **56 and 90 bags per hectare respectively**. The price of maize is **Ksh. 1200 per bag and Ksh. 300 per bag of potatoes**. Seed costs **Ksh 1350 per 10 kg maize and ksh 200 per 50 kg potatoes**. DAP fertilizer cost **ksh 1400 per 50 kg bag**. Labour cost **ksh 150 per man day**. MWANGI requires **10 bags of potatoes seed and 1 bag of maize to cover 0.3 hectares of land**.

- a) Draw up a **partial budget to determine** the effect of the change. ($1/2 \times 8 = 4$ marks)
 b) **State** the rational decision he should make basing on your calculations. (1 mark)

SECTION C

ANSWER ANY TWO QUESTIONS IN THIS SECTION.

24.) **Describe** the production of tomatoes **under** the following **subheadings**.

- a) Transplanting. (10 marks)
 b) **Field** management practices. (10 marks)
 25.a) **Explain** the following **terms** as used in product-product relationships.
 i) **Supplementary products**. (2 marks)
 ii) **Complimentary products**. (2 marks)
 b) **Explain seven factors** considered during farm planning. (14 marks)
 26.a) **Give five ways** of controlling water pollution. (10 marks)
 b) **Explain five problems** associated with credit in agricultural production. (10 marks)

PAPER 2

SECTION A 30 MARKS

ANSWER ALL THE QUESTIONS

- 1.) State **four** factors that affect the distribution of livestock in Kenya. (2 marks)
 2.) **Give** examples of each of the following types of livestock animals.
 a) Dairy cattle (1/2 mark)
 b) Dual purpose sheep (1/2 mark)
 c) Hair goat. (1/2 mark)
 d) Dual purpose cattle (1/2 mark)
 3.) Name **four** livestock tools that are used together in complimentary manner. (2 marks)
 4.) **State** the function of a spoke shave. (1 mark)
 5.) State **four** reasons for maintenance practices carried out on masonry tools. (2 marks)
 6.) Outline **four** reasons for keeping livestock healthy. (2 marks)
 7.) State **four** harmful effects of parasites (2 marks)
 8.) Name **three** methods of selection in livestock production. (1 1/2 mark)
 9.) State **two** signs of parturition in pigs. (2 marks)
 10.) **Differentiate** between **steaming up** and **flushing**. (1 mark)
 11.) State **four disadvantages** of using a spray race in controlling parasites in livestock. (2 marks)
 12.) **Outline four** structural requirements of a calf pen. (2 marks)
 13.) **Differentiate** between **cropping** and **harvesting** as used in **aquaculture**. (1 marks)
 14.) List **four factors** that influence the choice of poultry rearing system. (2 marks)

15) **Outline four** rules observed when milking.

(2 marks)

16) **List four** importance of lubrication system in a tractor.

(2 marks)

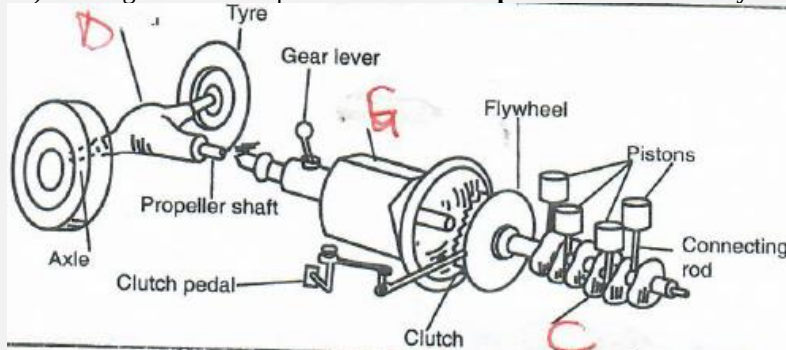
17) **Outline three advantages** of setting up a biogas unit in a dairy unit.

(1 ½ marks)

SECTION B-20marks

18) Use the **Pearson square method** to formulate Duck mash 28% DCP using maize DCP 11% and fishmeal 35% DCP. Show all your working (5 marks)

19) The diagram below represents the **tractor power transmission system**. Study it carefully and answer the questions that follow



a) **Name three aspects** that ensure maximum traction in a tractor

(1 1/2 marks)

b) Name parts

(1 ½ marks)

C

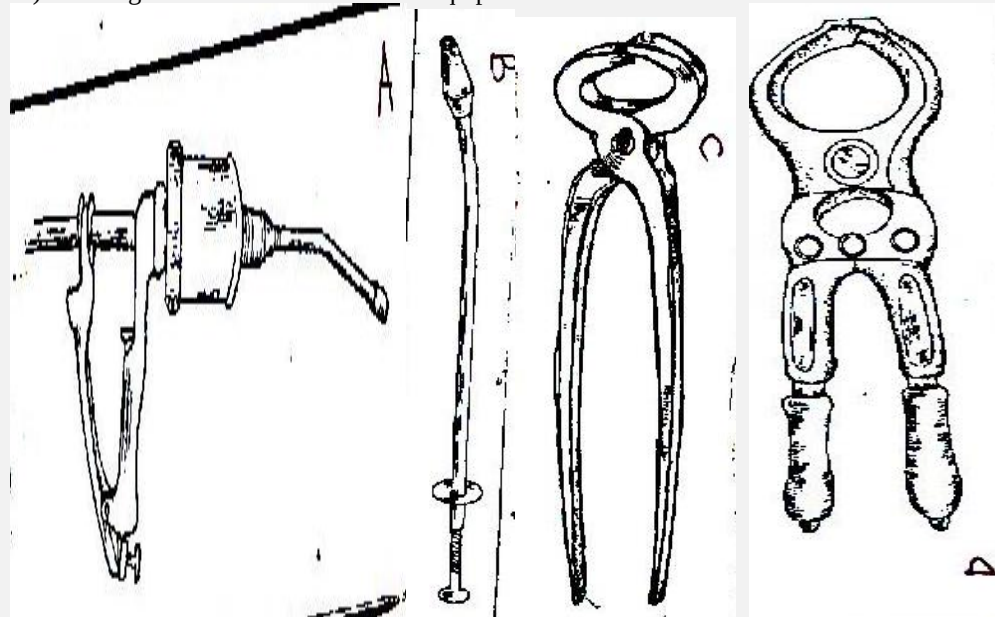
D-

G-

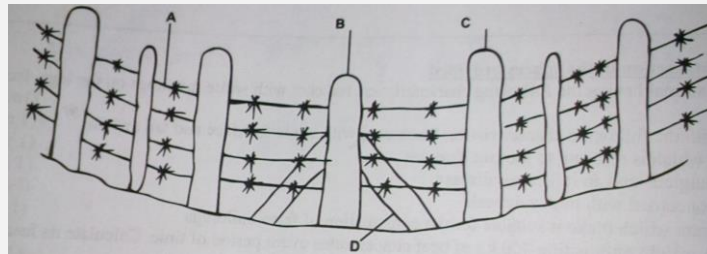
C) **State two** functions of part D

(2 marks)

20) The diagrams below shows various equipment used in **livestock health**



- a) **Name** equipment A,B,C,D (2 marks)
 b) State the function of D. (1 mark)
 c) Give the **functional difference** between A and B (2 Marks)
 21) The diagram below represents an **important farm structure**. Study it and the answer. Study it and answer the questions that follow.



- (i) Identify the (1mk) farm structure
 (ii) Label the parts. A,B,C,D (2mks)
 (iii) **State two** maintenance practices carried out in the structure above. (2mks)

SECTION C

Answer **any 2** questions only

- 22) a) **State and explain five factors** that increase the digestibility of feeds (10 marks)
 b) **State and explain** five ways of preventing diseases in livestock. (10 marks)
 23) a) **Describe five** structural requirements of modern produce store. (10 marks)
 b) **Describe** the sequence of events that take place in a 4-stroke petrol engine in one cycle starting with induction, compression, power and exhaust stroke. (10 marks)
 24) a) **Give four** reasons for inbreeding in livestock (4 marks)
 b) **Outline four** characteristics of eggs for incubation (4 marks)
 c) State the use of each of the following milky materials and equipment.
 (i) Two towels per cow (1mks)
 (ii) Filter pads (1mks)
 (iii) Milking jelly (1mks)
 (iv) Strip cup (1mks)
 d) **Describe** bloat under the following subheadings
 (i) Animals affected. (2 marks)
 (ii) Causal agent. (2 marks)
 (iii) Symptoms. (2 marks)
 (iv) Control and treatment. (2 marks)

MODEL TEST PAPER 7

PAPER 1

SECTION A: (30 MARKS)

Answer all the questions in this section in the spaces provided.

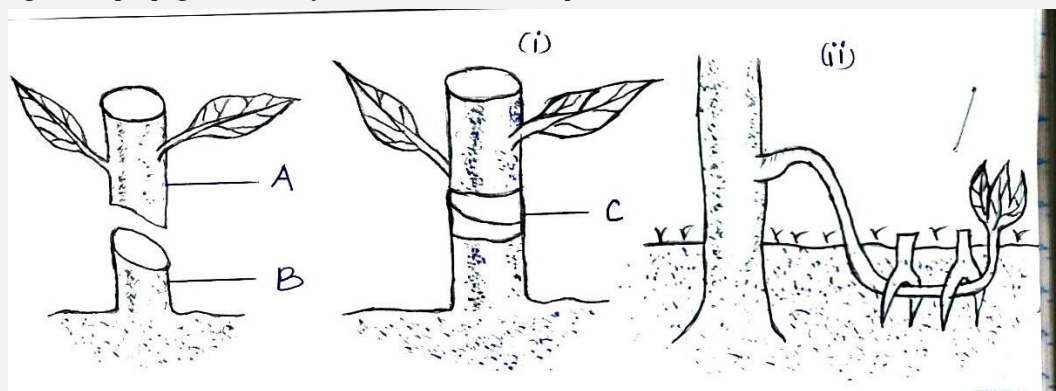
- Name four methods of farming. (2mks)
- State four pieces of information contained in the land title deeds. (2mks)
- a) State two factors that determine the stage at which the crop is harvested. (1mk)
 b) Outline four factors considered when designing a crop rotation programme. (2mks)
- a) Define sub soiling. (1mk)
 b) Give two causes of hardpans. (1mk)
- State four uses of phosphorus in the growth and development of crops. (2mks)
- Give four aspects of rainfall that a farmer may consider when deciding on type of crop to grow. (2mks)
- List four sources of organic matter in soil. (2mks)

8. Give four reasons why seed selection is important in the establishment of crops. (2mks)
9. a) Give two aspects of a good farm record. (1mk)
- b) What are two situations in farming where opportunity cost is zero. (1mk)
10. State four advantages of timely planting. (2mks)
11. Name three types of leguminous fodder crops. (1 ½ mks)
12. Give three maintenance practices for trees in agroforestry. (1 ½ mks)
13. State four problems that farmers are likely to face when marketing their produce. (2mks)
14. Differentiate between partial budget and complete budget. (2mks)
15. Give four physical agents of weathering. (2mks)

SECTION B:(20 MARKS)

Answer all the questions in this section

16. a) A farmer is to carry out farming in a soil that requires 50kg P_2O_5 and has a fertilizer labelled 20:30:0. How much fertilizer does he require? (4mks)
- b) What is the percentage of P_2O_5 in the fertilizer. (1mk)
17. The diagram labelled A, B, (i) and (ii) below illustrates materials and methods of vegetative propagation. Study them and answer the questions that follow.



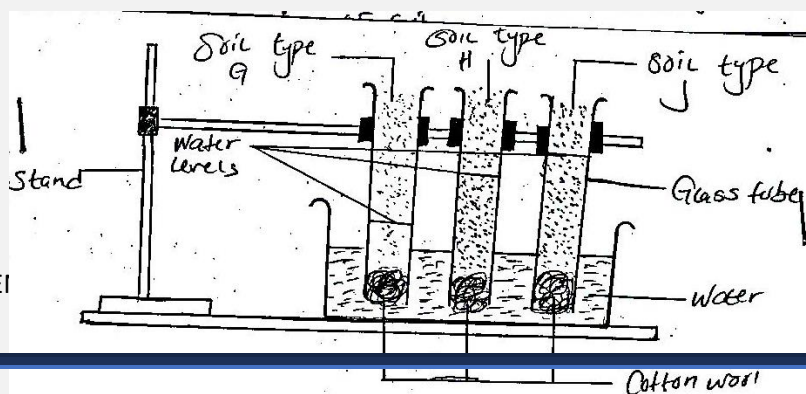
- a) Name the parts labelled A and B. (2mks)
- b) Give the reason for carrying out practice C in diagram (i) (1mk)
- c) Name the methods of propagation illustrated in diagrams (i) and (ii) (2mks)

18. Diagrams G and H below show some weeds.



- a) Identify the weeds G and H (2mks)
- b) State one economic importance of weed H. (1mk)
- c) Why is it difficult to control weed G. (1mk)
- d) Name an example of a systemic herbicide which can be used to control weed G. (1mk)

19. The diagram below shows an experiment set up using soil types G, H and J to investigate a certain property of soil.



- a) Identify the property being investigated. (1mk)
- b) Name the three soil types. (3mks)
- G -
- H -
- J -
- c) What conclusion can be drawn from the above experiment. (1mk)

SECTION C:(40 MARKS)

Answer any two questions in this section

20. a) Name four types of soil erosion by water. (4mks)
- b) Explain eight factors influencing soil erosion. (16mks)

21. The following is a farm record which Mr. Kamau had kept on 30th June 2008. Study it carefully and answer the questions that follow.

	Ksh.
Cash in hand	20,000
Cash at bank	66,000
Buildings	50,000
Disc plough	16,000
Debtors	16,000
Working tools	12,000
Bank overdraft	24,000
Creditors	20,000
Loan	50,000
Cattle	40,000
Land	80,000

- i) Prepare the balance sheet from the above information for Mr. Kamau's farm. (6mks)
- ii) State whether Mr. Kamau's farm business is solvent or insolvent. (1mk)
- iii) State three benefits of the balance sheet to Mr. Kamau. (3mks)
- b) Explain five risks and uncertainties in farming. (10mks)
- 22.a) Describe the cultural methods of weeds control in crop production. (10mks)
- b) Describe the harmful effects of pests on crops (10mks)

PAPER 2

SECTION A:(30 MARKS)

Answer all the questions in this section

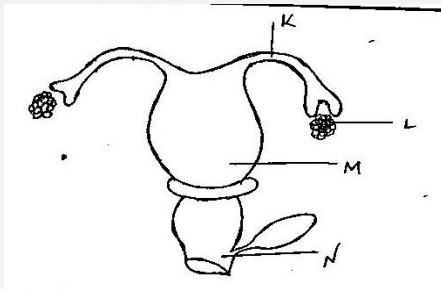
- State four importance of keeping livestock healthy. (2mks)
- Distinguish between the following practices as used in livestock production.
 - Crutching and ringing in sheep management. (2mks)
 - Steaming up and flushing. (2mks)
- State three observable features that help to differentiate the Dromedary camel from the Bactrian camel. (1 ½ mks)
- Name two functions of the crop in digestive system of chicken. (1mk)
- a) Give three reasons for candling eggs in poultry production. (1 ½ mks)
- b) State three qualities of the shell that should be considered when selecting eggs for incubation. (1 ½ mks)
- Give four predisposing factors to mastitis in dairy cattle (2mks)
- Outline four mechanical methods of controlling ticks. (2mks)
- State four structural features of ideal calf pen. (2mks)
- Give two problems associated with tractor hire services. (1mk)
- Give the functions of following farm tools and equipments. (2mks)
 - Sickle

- ii) Strip cup
 - iii) Claw hammer
 - iv) Mason's trowel
11. State four properties of a good vaccine. (2mks)
12. Give four factors that affect digestibility of feeds in animals. (2mks)
13. Outline four farm structures that are necessary for handling dairy animals. (2mks)
14. Give three ways in which farmers market beef cattle in Kenya. (1 ½ mks)
15. List four farm implements attached to power take off (PTO) (2mks)

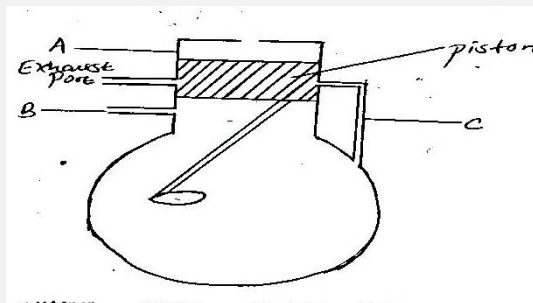
SECTION B:(20MKS)

Answer all the questions in this section in the spaces provided.

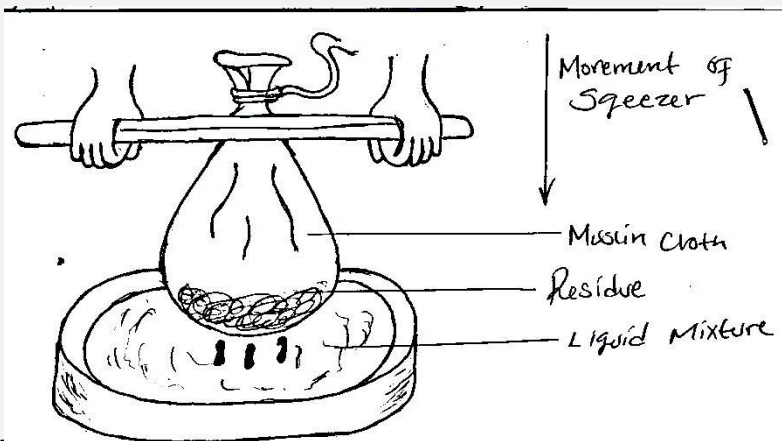
16. a) A dairy farmer is required to prepare 100kg of dairy meal containing 20% Digestible crude protein (D.C.P) using the Pearson's square method, calculate the quantity of soya bean (40% DCP) and rice (16% DCP) the farmer requires for the dairy meal (show your working) (4mks)
- b) Name one other method that can be used to compute livestock ration. (1mk)
17. The diagram below shows the female reproductive system of a cow. Use it to answer the question that follow.



- a) Name the parts labelled K, L, M and N. (2mks)
- b) Give the functions of the parts labelled K, L and M. (3mks)
18. The diagram below illustrates a two – stroke



- i) Name parts A, B and C. (3mks)
- ii) Identify the stroke shown above. (1mk)
- iii) Indicate on the diagram using letter X where compression of fuel Mixture occur. (1mk)
19. Below is an illustration of a method used by farmers to extract honey from the combs.



- a) Identify the method. (1mk)
- b) Name any other methods that can be used to process honey. (1 mk)
- c) Describe briefly how the method illustrated above is carried out on the farm. (3mks)

SECTION C:(40 MARKS)

Answer any two questions in this section in the spaces provided.

- 20. a) Describe parts and functions of a plunge dips. (14mks)
b) State six advantages of live fences. (6mks)
- 21. a) Describe the management of one day old chicks in a brooder until they are eight weeks. (12mks)
b) Explain four causes of cannibalism in poultry production. (8mks)
- 22. a) Describe newcastle disease under the following sub – headings.
i) Causal organism (1mk)
ii) Signs of infection (7mks)
iii) Control measures. (2mks)
- b) Explain five factors to consider when selecting a breeding stock. (10mks)

MODEL TEST PAPER 8

PAPER 1

SECTION A (30 marks)

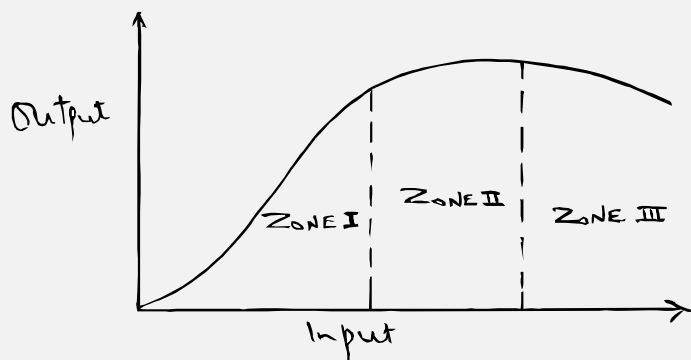
Answer ALL questions in this section on the spaces provided.

1. State two reasons for root trimming in a tree nursery. (1 mark)
2. Give two forms in which nitrogen is absorbed from the soil by plants. (1 mark)
3. State four financial documents that should be kept on farm. (2 marks)
4. State four ways in which burning of vegetation may lead to loss of soil fertility. (2 marks)
5. Give two for locating a nursery bed in a well sheltered place. (1 mark)
6. List down four methods of treating water for use on the farm. (2 marks)
7. Give two examples for each of the following categories of water pipes.
a. Metal pipes. (1 mark)
b. Hose pipes. (1 mark)
8. Give two ways in which cover crops help to conserve water in the soil. (1 mark)
9. The following are different types of costs used during production :
Rent ; wages of casual labour ; fuel costs; salaries of permanent labour.
Categorize the cost in to :
a. Fixed costs. (1 mark)
b. Variable costs. (1 mark)
10. What is the function of each of the following during water treatment.
a. Soda ash (1 mark)
b. Alum (1 mark)
c. Chlorine (1 mark)
11. State four importance of agroforestry in the farm. (2 marks)
12. a. Give two conditions in agricultural production under which opportunity cost is zero. (1 mark)
b. Apart from opportunity cost, mention two other basic concepts of economics. (1 mark)
13. State two ways in which agriculture contributes directly to industrial development. (1 mark)
14. State four advantages of mixed farming. (2 marks)
15. Give four importance of ridging in crop production. (2 marks)
16. Name four elements whose deficiency causes chlorosis in plants. (2 marks)
17. Differentiate between consumable goods inventory and permanent goods inventory. (2 marks)

SECTION B (20 marks)

Answer All the questions.

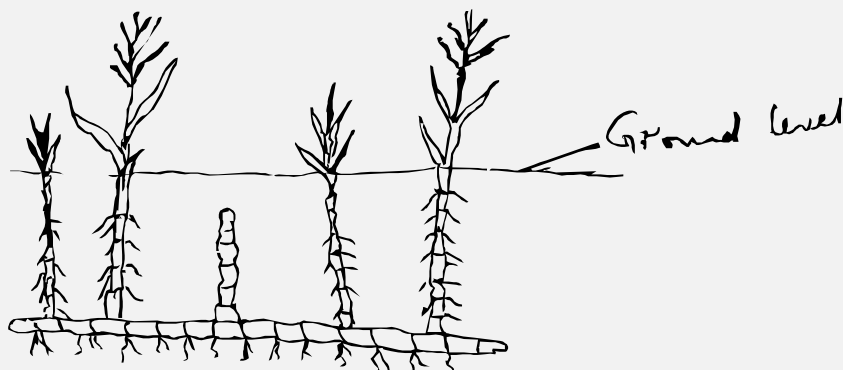
18. Below is a graphical representation of an economic law.



- a. Identify the law illustrated above. (1 mark)
- b. Explain what happens in each of the zones in relation to output.
Zone I
Zone II
Zone III (3 marks)
- c. Which of the three is the rational zone of production. (1 mark)

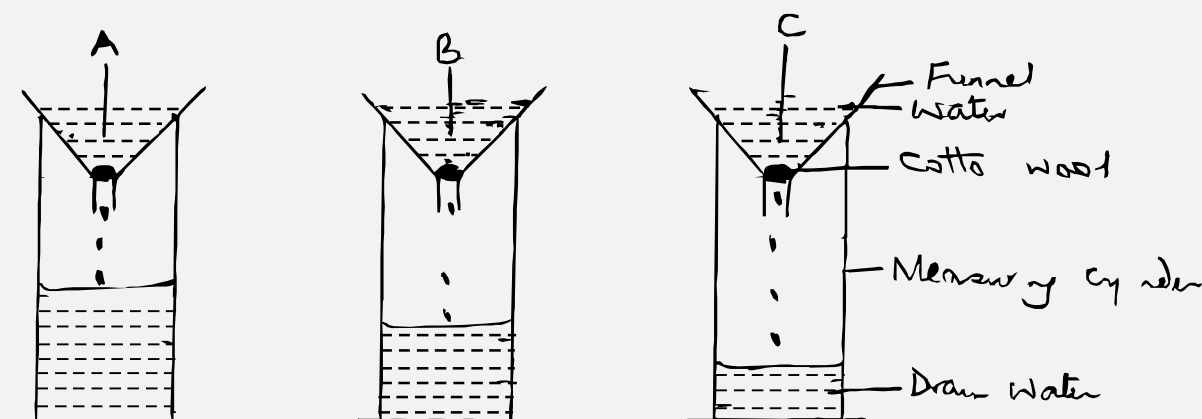
19. The diagram below illustrates a common weed that inhabits both arable and pasture land.

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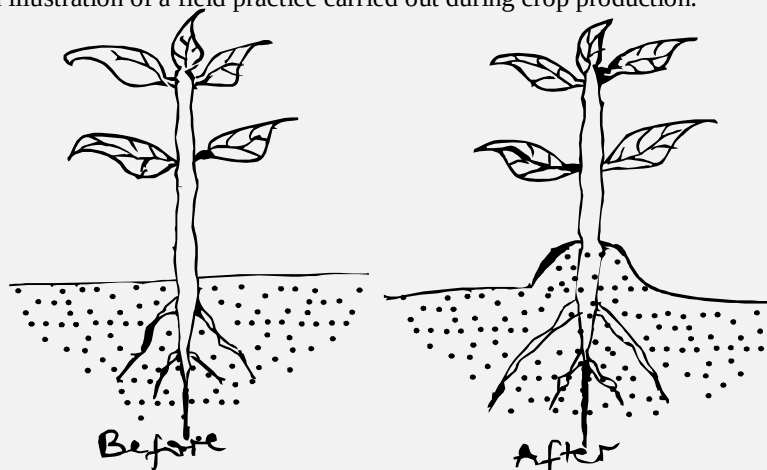
- Identify the weed illustrated above. (1 mark)
- Classify the above weed according to :
 - Growth cycle (1 mark)
 - Plant morphology. (1 mark)
- Give two appropriate methods of controlling the above weed. (2 marks)

20. The diagram below illustrates an experiment on soil. Study it carefully and answer the questions that follow.



- State the aim of the experiment. (1 mark)
- If the amount of water added in each soil was the same, identify the soil samples A, B and C. (3 marks)
- State two ways in which the soil structure of the soil sample C can be improved. (2 marks)

21. Below is an illustration of a field practice carried out during crop production.



- Identify the practice illustrated above. (1 mark)
- Citing a crop in each case, give three importance of the above practice in crop production. (3 marks)

SECTION C 40 marks)

Answer two questions from this section. All questions carry equal marks.

22. a. Explain five advantages of mulching in crop production. (5 marks)
b. Outline five activities that may be undertaken in organic farming. (5 marks)
c. Give five benefits of using vegetative propagation in production of citrus. (5 marks)
d. Describe five management practices carried out on pasture. (5 marks)
23. a. Describe eight farming practices that encourage soil erosion. (8 marks)
b. Explain five groups of pesticides based on the mode of action. (5 marks)
c. i. Describe procedure of harvesting pyrethrum. (4 marks)
ii. Explain precautions that should be observed during the harvesting of pyrethrum. (3 marks)
24. a. Describe eight principles that govern the running of co-operatives. (8 marks)
b. Describe carrot production under the following sub-headings :
i. Field management practices. (5 marks)
ii. Harvesting. (2 marks)
c. Outline five practices used to control maize streak disease. (5 marks)

PAPER 2

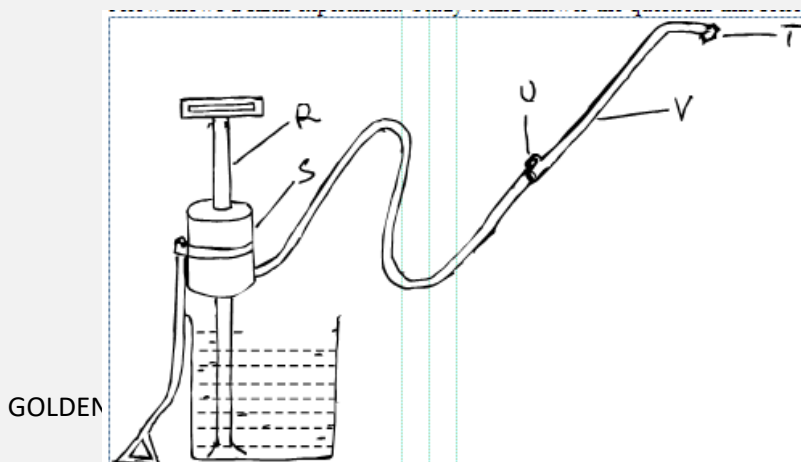
Answer ALL questions in this section on the spaces provided.

1. List four tools used in laying concrete blocks in construction of a wall. (2 marks)
2. Give two reasons why farmers prefer to keep hybrid birds to pure breeds. (1 mark)
3. Differentiate between notifiable disease and zoonotic disease. (2 marks)
4. Outline any four signs of infestation by external parasites in livestock. (2 marks)
5. State four factors that determine nutritional requirements in dairy goats. (2 marks)
6. Give four advantages of artificial insemination. (2 marks)
7. Outline four importance of castration in male animals. (2 marks)
8. State the function of each of the following tools in bee handling.
a. Bee brush (1 mark)
b. Hive tool (1 mark)
c. Hive knife (1 mark)
9. State four properties of concrete that make it suitable for constructing farm buildings. (2 marks)
10. Give two reasons why maintenance of farm structure is important. (1 mark)
11. Name two non-contagious protozoa diseases of livestock. (1 mark)
12. State four signs of foot rot in sheep. (2 marks)
13. List four factors a farmer should consider when grading eggs for marketing. (2 marks)
14. Outline four routine management practices carried out in female calves. (2 marks)
15. Give four functions of lubricants in an engine. (2 marks)
16. Name the dairy breed of cattle that :
i. Produces the highest amount of milk per lactation period. (1 mark)
ii. Produces milk with lowest butter fat content.
17. State two socio-cultural uses of cattle in Kenya. (1 mark)

Section B (20 marks)

Answer ALL the questions on the spaces provided.

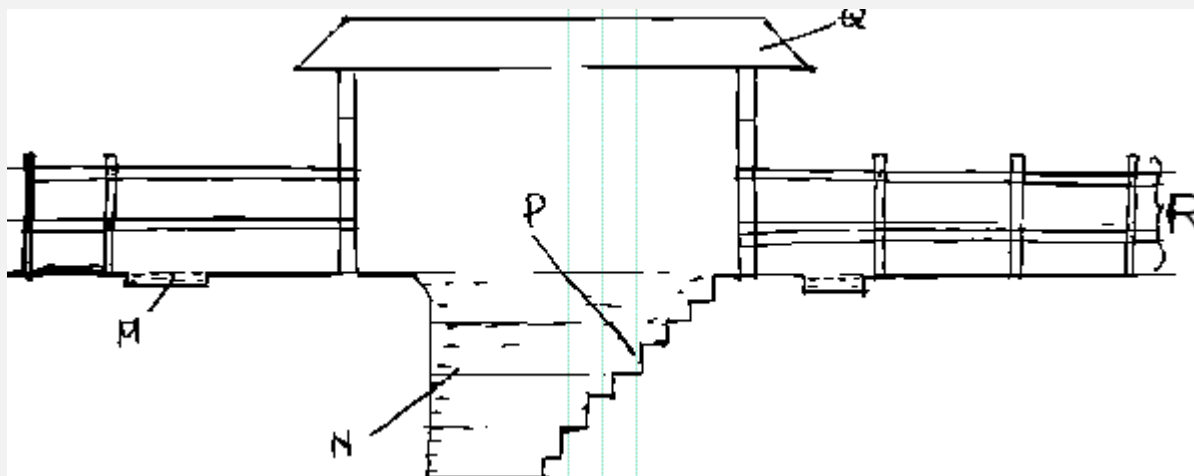
18. The diagram below shows a farm experiment. Study it and answer the questions that follow.



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- a. Identify the experiment. (1 mark)
- b. Give the appropriate use of the equipment above. (1 mark)
- c. Name the part labeled R. (1 mark)
- d. What is the function of the part labeled T.? (1 mark)
- e. Give one maintenance practice of the equipment.
19. Compute 100Kg ration for layers containing 16% DCP using a mixture of maize 8 % DCP and sorghum 8.8 % in the ratio 1:1 and cotton seed cake 24 % DCP. (6 marks)

20. The diagram below shows a livestock handling structure. Study it and answer the questions that follow.



- a. Identify the structure above. (1 mark)
- b. Identify the parts labeled M, N, P and Q. (4 marks)
- c. Give one importance of each of the parts labeled M and P. (2 marks)
- d. Give two structural features of the part labeled R. (2 marks)

Section C (40 marks)

Answer any two questions in this section

21. a. Describe short term tractor servicing. (10 marks)
- b. Outline the practices carried out in a deep litter poultry rearing system to control diseases and parasites. (10 marks)
22. a. What are predisposing factors of a disease? (1 mark)
- b. Describe mastitis under the following sub-headings
 - i. Causal organism. (1 mark)
 - ii. Predisposing factors (4 marks)
 - iii. Symptoms. (4 marks)
- c. Describe factors that may necessitate culling in livestock production. (10 marks)
23. a. Describe the management practices that a farmer should carry out to increase milk production in the a yielding herd. (10 marks)
- b. Outline general methods of disease control. (5 marks)

c. State and explain five parts of a zero grazing unit.

(5 marks)

MODEL TEST PAPER 9

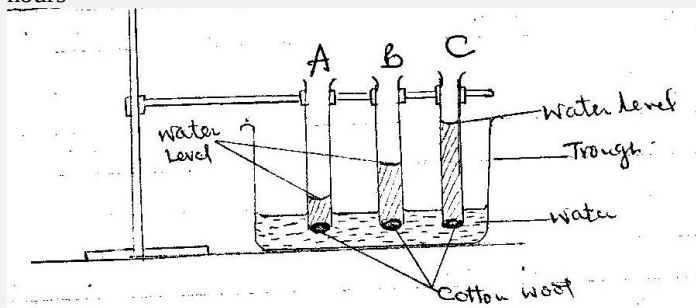
PAPER 1

SECTION A (30 MARKS)

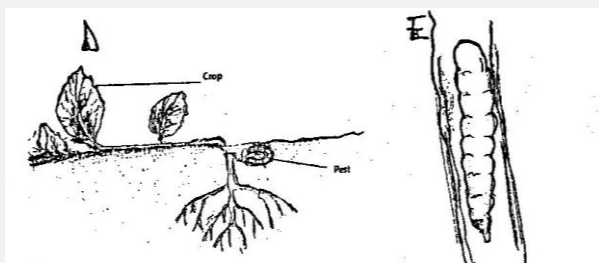
1. What is plantation farming system (1mk)
2. State two effects of high level of education and technology on agricultural production (1mk)
3. Name two ways in which soil Ph is raised (1mk)
4. State two means by which water is conveyed on farms (1mk)
5. Give four factors that affect the quality of farm yard manure (2mks)
6. Mention two conditions under which the opportunity cost is zero (1mk)
7. List any three methods which can be used to detect nutrient deficiency in crops (1 ½ mks)
8. State four reasons for processing agricultural produce (2mks)
9. Give four factors considered when grading tomatoes for fresh market (2mks)
10. State four reasons why rough lemon is preferred as a root stock in the grafting of citrus (2mks)
11. Highlight four objectives of land settlement which have been undertaken in Kenya (2mks)
12. Mention four types of soil erosion by water (2mks)
13. Name two insect pests that attack sorghum in the field (1mk)
14. Give two benefits of possessing a land title deed to a farmer (1mk)
15. List four variable costs in coffee production (2mks)
16. Outline four financial documents (2mks)
17. State four reasons why timely weed control is recommended in crop production (2mks)
18. Distinguish between over sowing and under sowing as used in crop production (1mk)
19. Give three ways in which forage crops are classified (1 ½ mks)
- b) What is intensive hedgerow as used in agro forestry? (1mk)

SECTION B (20MKS)

20. The diagram below shows an experiment set up using garden soil samples types A,B and C. The observations were made after 24 hours

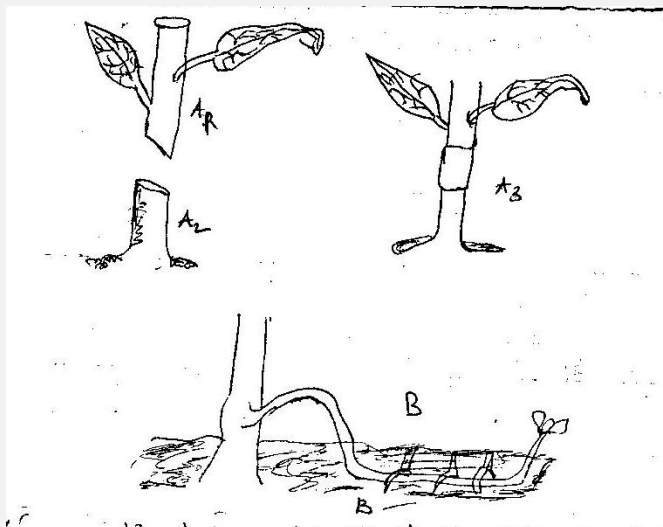


- i) What is the experiment above designed to study (1mk)
 - ii) Name the three soil types (1 ½ mks)
 - iii) State the characteristic texture of soil types (1 mk)
 - iv) Give three ways a farmer would improve the structure of soil type C (1 ½ mks)
21. The diagram below D and E illustrates some field pests. Study them carefully and answer the questions that follow



- i) Identify the pests D and E (1mk)
- ii) State two damages caused by pest E to a crop of maize (2mks)
- iii) Give two cultural practices carried out to control the pest in E (2mks)

22. The diagrams labeled A1, A2, A3 and B below illustrate materials and methods of vegetative propagation. Study them and answer the questions that follow



- Identify the parts labeled A1 and A2 (1mk)
- State the methods of propagation illustrated in diagrams A3 and B (2mks)
- Name two other methods of A3 (2mks)

23. A farmer can combine dairy meal and homemade feeds in feeding dairy cattle. Study the table below and answer the questions that follow

Dairy MEAL (KG)	Homemade feed (kg)	Marginal rate of substitution
1	48	O
2	39	V
3	32	7
4	27	W
5	23	4
6	21	X
7	20	1
8	19	Y

- Given the information above. Calculate the marginal rate of substitution and give values for V,W,X,Y (2mks)
- Suppose, when the price of a bag of maize is 2000ksh per bag. 20 bags are demanded but when the price changed to Kshs 1800 per bag, 22 bags are demanded. Calculate the Elasticity of demand. (Show your working) (2mks)
- State the type of elasticity of demand in (ii) above (1mk)

SECTION C (40MARKS)

24a. Describe the field production of Napier grass under the following sub headings

- Planting (4mks)
- Fertilizer and manure application (3mks)
- Utilization (3mks)

- Describe qualities of a mother plant which should be considered when selecting vegetative materials for planting (5mks)
- Describe advantages of tillage as a method of weed control (5mks)

25a. Describe various agencies involved in marketing agricultural products (5mks)

- Explain five factors that influence soil productivity (5mks)
 - Describe the procedure followed when transplanting tree seedlings from a bare root nursery bed (10mks)
- 26a. Explain factors that should be considered when selecting seeds for planting (10mks)
- Describe the procedure of harvesting pyrethrum (4mks)
 - Describe six advantages of minimum tillage (6mks)

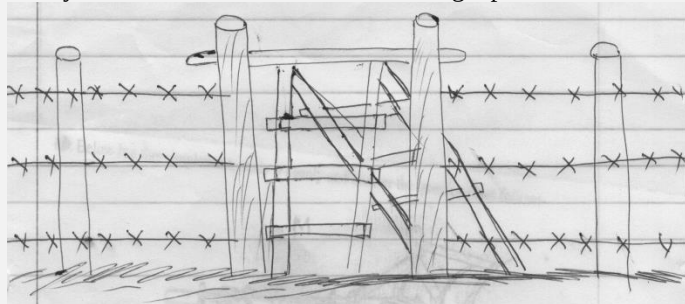
PAPER 2

SECTION A (30MKS)

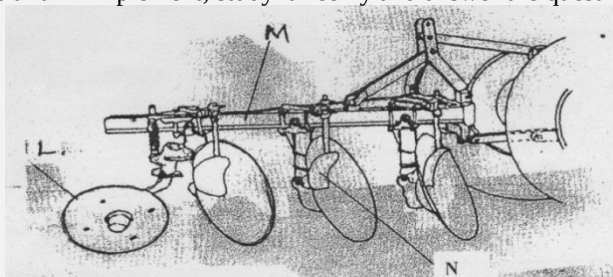
1. Name two tools used for cutting galvanized iron pipes (1mk)
2. List two pork breeds of pigs (1mk)
3. Outline four routes through which disease causing organisms get into the body of an animal (2mks)
4. Name two developmental stages of a liver fluke in water snail (1mk)
5. Name four sources of carbohydrates in livestock nutrition (2mks)
6. Give four importance of docking in sheep management (2mks)
7. What does the term Epistaxis mean in livestock improvement (1mk)
8. Give two disadvantages of a barbed wire fence when used in paddocking (1mk)
9. Outline four symptoms of attack by coccidiosis disease in kids (2mks)
10. Name four causes of stress in a flock of layers (2mks)
11. State four advantages of natural feeding in calf rearing (2mks)
12. Give four implements that obtain power from power take off shaft of the tractor (2mks)
13. State four factors that stimulate milk letdown in a lactating cow (2mks)
14. List four desirable qualities of a livestock ration (2mks)
15. Give three methods of applying acaricides on animals (1 ½ mks)
16. List four components of power transmission system (2mks)
17. Give three signs of parturition in a doe (1 ½ mks)
18. State two factors that lower the quality of concrete (1mk)
19. State two uses for which wind power is harnessed (1mk)

SECTION B

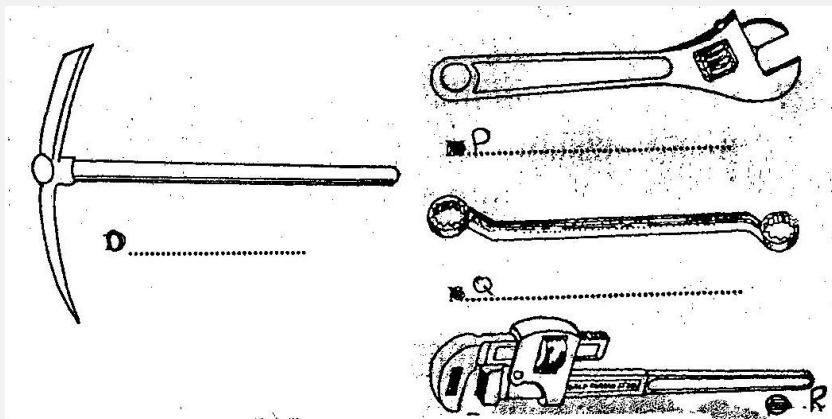
20. Study the farm structure below illustrating a pass



21. a) Identify the type of pass illustrated below (1mk)
b. Distinguish between a pass and a gate (1mk)
c. List four types of tools and equipments used in the establishment of the structure above (2mks)
d) one maintenance carried out on the structure (1mk)
21. Below is a farm implement, study it keenly and answer the questions that follow



- a. Name the farm implement drawn above (1mk)
b. Identify the parts labeled L and M above
c. State the **function** of the part labeled M (1mk)
d. State two field conditions under which the implement can work better than others (1mk)
- 22i) Using Pearson's square method, calculate how much maize meal (8% DCP) and soya bean meal (43%DCP) would be required to
- i. prepare a 100kg ration of the mixture containing 15%DCP (3mks)
 - ii) State four factors considered in selecting feedstuffs for computation (2mks)
23. Below are farm tools, study them and answer the questions that follow



- a) Identify the tools O,P,Q,R (2mks)
- b) State the use of tool R (1mk)
- c) State four reasons for maintaining farm tools and equipments (2mks)

SECTION C

Attempt two questions from this section

24A) Describe East Coast fever under the following sub-headings

- i) Causal organisms (1mk)
- ii) Symptoms of attack (5mks)
- iii) Control measures (4mks)

b. Explain management of a cow during parturition (10mks)

25a. Describe the daily maintenance practices of a tractor (10mks)

b. Describe the characteristics of indigenous cattle that make them more suited to survive in marginal areas (10mks)

26a) Explain the role of livestock industry in Kenya's Economy (10mks)

- a) Explain the principles considered when controlling Endo-parasites (5mks)
- b) Describe the procedure of harvesting honey in a Kenya Top Bar Hive (5mks)

MODEL TEST PAPER 10

PAPER 1

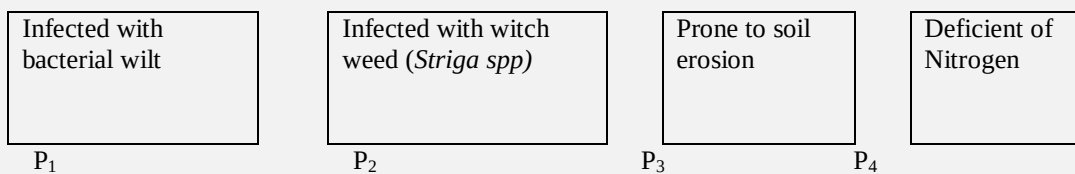
Answer ALL questions in the spaces provided

1. Name two forms of collective land tenure systems in Kenya. (1mk)
2. State three functions of plastic materials when used as mulches in crop production. (1½mks)
3. Give four reasons for keeping breeding records in a dairy farm. (2mks)
4. Give four importance of proper soil aeration in crop growth. (2mks)
5. State two characteristics of *katumani* maize variety which make it suitable for production in dry areas. (1mk)
6. Give four advantages of intensive farming. (2mks)
7. Mention any three reasons for processing agricultural produce. (1½mks)
8. List four variable costs in coffee production. (2mks)
9. Distinguish between nitrogen fixation and phosphorous fixation. (1mk)
10. State four characteristics of a good vegetable seedling. (2mks)
11. List four methods of harvesting trees in agroforestry. (2mks)
12. What is opportunity cost? (1mk)
13. State four ways of determining the rate of payment of casual labour. (2mks)
14. Give four factors that determine the time of planting annual crops. (2mks)
15. Name three crop diseases caused by fungi. (1½mks)
16. State four factors that determine the quality of hay. (2mks)
17. Mention three limitations of chemical weed control. (1½mks)
18. Give four farming practices that are used to achieve minimum tillage. (2mks)

SECTION B (20 Marks)

Answer ALL questions in this section in the spaces provided

19. A farmer has four plots of land P₁, P₂, P₃ and P₄ as shown on the diagram below. Each of these plots has an agronomic problem indicted. Study them carefully then answer the questions that follow.



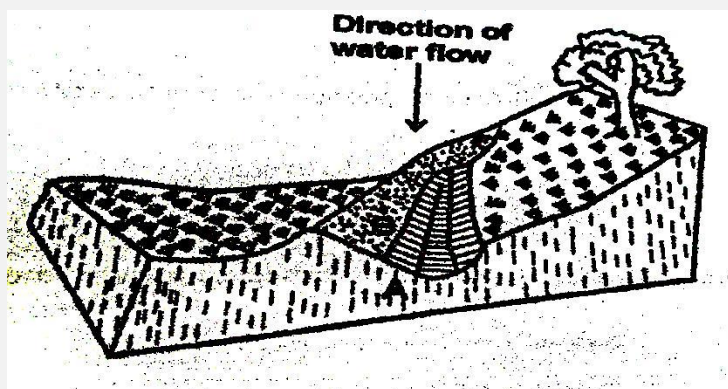
The farmer intends to grow Irish potatoes, peas, maize and Rhodes grass.

- i) Match each crop with the most appropriate plot, for the first year of the rotation system. (1mk)
- ii) Outline four factors considered when designing a crop rotation program. (4mk)
- iii) Highlight one way in which crop rotation controls weeds. (1mk)
- iv) State one reason for including grass leys in the rotation programme. (1mk)
20. a) State the law of diminishing returns in a production process. (1mk)
- b) Use the information in the table below to answer the questions that follow.

Fertilizer input	Maize yields (bags)	Marginal production (bags)
0	50	12
1	62	12
2	66	4
3	68	2
4	69	1
5	69	0

The price of fertilizer is ksh1500/- per unit and the price of maize is ksh1200/-per bag.

- i) At what unit of fertilizer input should the farmer be advised to stop applying any more fertilizer to the maize? (1mk)
 - ii) Give a reason for your answer in (i) (b) above. (1mk)
 - iii) Calculate the marginal return at the point of the optimum production. (2mks)
21. The illustration below shows a newly constructed cut-off drain. Study it carefully and use it to attempt the questions that follow.



- i) Identify the parts labelled A and B. (1mk)
- ii) How can the part labelled B be stabilized? (1mk)
- iii) State two factors that would determine the width and depth of the cut-off drain. (2mks)
- iv) Mention two ways to maintain a cut-off drain. (1mks)

22. A plot of land measures 6.6m long by 3.6m wide. The plot is prepared for planting cabbages at a spacing of 60cm x 60cm. The outermost rows start 30cm from the edges all-round the plot. The rows move along the length of the plot. Showing your working, calculate ;

- i) The number of rows falling on the width side of the plot. (2mk)
- ii) The number of cabbage seedlings that should be planted in the plot. (3mks)

SECTION C (40Marks)

Answer any two questions in the spaces provided

23. a) With relevant example classify weeds according to growth cycle and morphology. (5mks)
- b) Describe the procedure followed in soil sampling. (6mks)
- c) Outline **nine** factors considered when choosing a farming system. (9mks)
24. a) Describe the harvesting of bulb onions. (5mks)
- b) Explain **five** roles of drainage as a method of land reclamation. (10mks)

- c) On 1st January 2017 Gwitu farm started farm operations with ksh 30,000/- cash.

During the month, the farm made the following transactions. Study the transactions and prepare a cash analysis for Gwitu farm for the month of January 2017. (5mks)

Date	Transaction	Amount (ksh)
05/01/17	Livestock sales	8,000
08/01/17	Crop sales	50,000
15/01/17	Bought seeds for planting	7,500
20/01/17	Paid K.F.A for fertilizer	16,400
25/01/17	Bought livestock feeds	50,000
30/01/17	Paid wages for planting and weeding	56,000
31/01/17	Received cash from K.C.C for milk delivered	120,000
31/01/17	Paid transport charges for milk delivery	9,000

25. a) Give **five** roles of organic matter in the soil. (5mks)
- b) Outline **five** non-chemical methods of controlling storage pests. (5mks)
- c) Describe the procedure followed during preparation of silage using maize. (10mks)

PAPER 2

TIME: 2 HOURS

SECTION A (30 MARKS)

Answer **all** the questions in this section in the spaces provided.

1. Give **three** factors that affect the quality of honey. (1½ marks)
2. State **three** conditions that make a cow withhold milk during milking. (1½ marks)
3. State **four** signs of farrowing observed in pigs. (2 marks)
4. Give **three** factors that may make birds lay eggs on the floor in a deep litter. (1½ marks)
5. Give **three** characteristics of clean and high quality milk. (1½ mark)

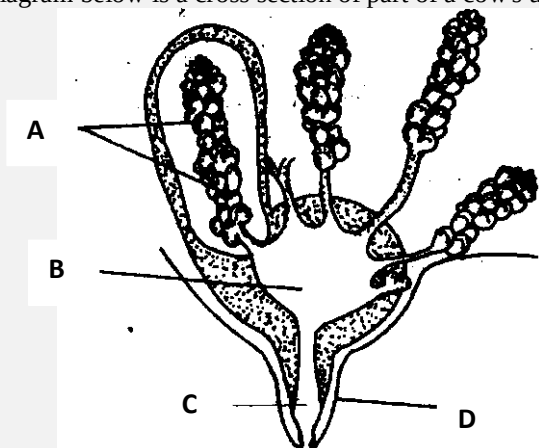
GOLDEN ELITE EDUCATIONAL CONSULTANCY P.O BOX 2345 KERICHO. FOR MARKING SCHEMES CALL OR TEXT MR CHEPKWONY ON 0724351706 OR EMAIL kipkemoicos@gmail.com

6. Outline **four** control measures of round worms. (2 marks)
7. State **four** signs of liver fluke infestation in cattle. (2 marks)
8. Give **four** reasons of breeding in cattle. (2 marks)
9. List **four** methods of preserving fish. (2 marks)
10. Name **two** classes of livestock feedstuffs. (1 mark)
11. Give **two** roles of ovaries in cows' reproductive system. (1 mark)
12. Give **two** reasons for flushing in sheep. (1 mark)
13. Give **four** reasons for treating timber used in construction of farm buildings. (2 marks)
14. State **four** factors that influence the daily water intake in an animal. (2 marks)
15. Name **two** functions of a clutch in the tractors transmission system. (1 mark)
16. Give **four** characteristics of a good calf pen. (2 marks)
17. State **four** factors that influence the choice of the poultry system of rearing to use. (2 marks)
18. State **four** routine management practices carried out on a replacement stock that is a heifer. (2 marks)

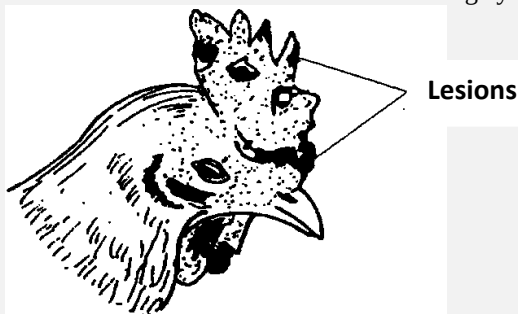
SECTION B (20 MARKS)

Answer **ALL** questions in this section in the spaces provided.

19. The diagram below is a cross section of part of a cow's udder.

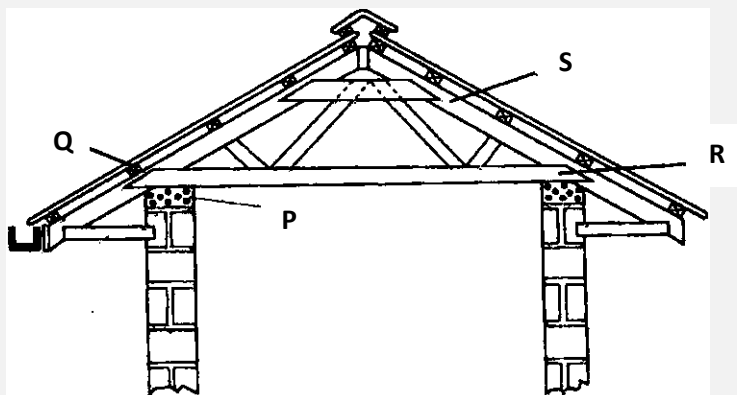


- a) Name the parts marked A, B, C and D. (2 marks)
 - b) Name **two** hormones that control milk let down in a dairy cow. (2 marks)
 - c) What is a dry cow therapy? (1 mark)
20. The diagram below shows the head of a chicken having symptoms of a poultry disease.

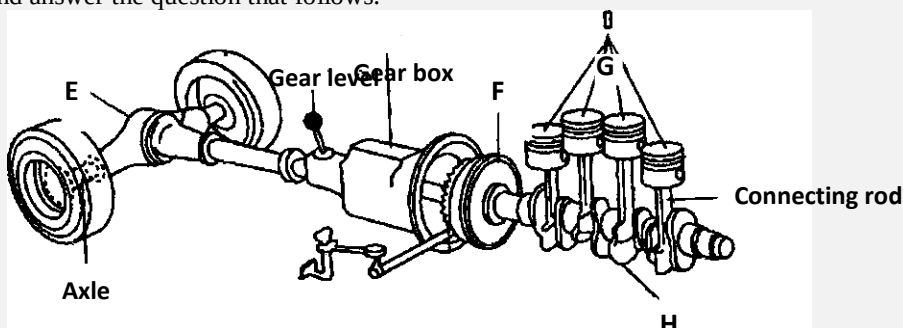


- a) Identify the disease. (1 mark)
- b) Give **two** reasons why the disease is of economic importance. (2 marks)
- c) Outline **two** methods of controlling the above disease. (2 marks)

21. The following diagram shows parts of a roof. Study it carefully and answer the questions that follow.



- a) Name the parts labeled P, Q, R and S. (2 marks)
 - b) State the functions of P and Q. (2 marks)
 - c) Give **two** chemical preservatives for treating timber before use in the construction of farm structures. (2 marks)
22. Below is a diagram of power transmission system of tractor engine. Study it and answer the question that follows.



- a) Name the parts labeled E, F, G and H. (2 marks)
- b) State the functions of E and H. (2 marks)

SECTION C (40 MARKS)

Answer **ANY TWO** questions from this section.

23. a) Describe **five** pre-disposing factors of livestock diseases. (5 marks)
- b) Describe **six** mechanical methods of controlling ticks. (6 marks)
- c) Describe the factors that affect milk composition of a cow. (9 marks)
24. a) State **five** advantages of farm mechanization. (5 marks)
- b) Explain **five** advantages of using a mould board plough in seed bed preparation. (5 marks)
- c) Describe the use of various materials in construction of a Kenya Top bar hive. (5 marks)
- d) Outline the care and maintenance of a tractor water cooling system. (5 marks)
25. a) State **four** physical characteristics of a dairy cow. (4 marks)
- b) State **nine** factors to consider when selecting livestock for breeding. (9 marks)
- c) Describe milk fever in dairy cattle under the following sub-headings.
 - i) The cause of the disease (1 mark)
 - ii) Symptoms of the disease (4 marks)
 - iii) Control measures

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ON 0724351706**

**Also available are the following assignments for all classes
(F1-4)**

- 1) Kiswahili**
- 2) Maths**

- 3) English
- 4) Chemistry
- 5) Home science
- 6) Physics
- 7) geography
- 8) Biology
- 9) History
- 10)
- 11)
- 12)

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